Multiscale Supervised Kernel Dictionary Learning for SAR Target Recognition

Lei Tao, *Student Member, IEEE*, Xue Jiang, *Senior Member, IEEE*, Xingzhao Liu, *Member, IEEE*, Zhou Li, and Zhixin Zhou

***Abstract*— In this article, a supervised nonlinear dictionary learning (DL) method, called multiscale supervised kernel DL (MSK-DL), is proposed for target recognition in synthetic aperture radar (SAR) images. We use Frost filters with different parameters to extract an SAR image's multiscale features for data augmentation and noise suppression. In order to reduce the computation cost, the dimension of each scale feature is reduced by principal component analysis (PCA). Instead of the widely used linear DL, we learn multiple nonlinear dictionaries to capture the nonlinear structure of data by introducing the dimension-reduced features into the nonlinear reconstruction error terms. A classification model, which is defined as a discriminative classification error term, is learned simultaneously. Hence, the objective function contains the nonlinear reconstruction error terms and a classification error term. Two optimization algorithms, called multiscale supervised kernel K-singular value decomposition (MSK-KSVD) and multiscale supervised incremental kernel DL (MSIK-DL), are proposed to compute the multidictionary and the classifier. Experiments on the moving and stationary target automatic recognition (MSTAR) data set are performed to evaluate the effectiveness of the two proposed algorithms. And the experimental results demonstrate that the proposed scheme outperforms some representative common machine learning strategies, state-of-the-art convolutional neural network (CNN) models and some representative DL methods, especially in terms of its robustness against training set size and noise.**

***Index Terms*— Kernel dictionary learning (DL), multiscale supervised incremental kernel DL (MSIK-DL), multiscale supervised kernel K-singular value decomposition (MSK-KSVD), multiscale features, synthetic aperture radar (SAR) target recognition.**

I. Introduction

DIFFERENT from optical imaging, synthetic aperture radar (SAR), with all-weather, day or night imaging capabilities, plays an important role in the domination of Earth observation, including military reconnaissance and civilian field. The issue of automatic target recognition (ATR) for SAR image interpretation has received considerable attention pervasively, however, it is still a challenging problem. Owing to phase coherence of the received backscattering returns of a moving SAR sensor, speckle noise presenting in an SAR image leads to the target degradation [1]. In addition, target signature variability, resulting from the properties of SAR sensors, operating conditions, and mechanism of a SAR imaging process, makes SAR ATR more difficult [2]. Early SAR ATR mainly depends on the template-matching scheme [3], which quantifies the similarity between the query and templates produced by the training SAR objects. Nevertheless, it seriously suffers from the changes in the SAR target pose and the presence of speckle noise. With the rapid development of computing power, convolutional neural networks (CNNs) [4], [5] have attracted renewed attention. In addition, machine learning algorithms, such as support vector machine (SVM) [6], k-nearest neighbor (k-NN) [7], and their extensions, also demonstrate satisfactory recognition performance for ATR.

Generally, machine learning-based target recognition is often divided into two steps, i.e., feature extraction and classification. Feature extraction is a crucial image-processing procedure to map the original data into an abstract space and hence a more representative and discriminative feature can be obtained to describe the target. In terms of the representation ability and extraction form, the features are categorized into global features and local features. The global features are often extracted by utilizing certain mathematical transformation for the whole pixels of the images, such as principal component analysis (PCA) [8], [9] and linear discriminative analysis (LDA) [9], [10]. In contrast, local features, such as scale-invariant feature transform (SIFT) [11], [12], histogram of oriented gradients (HOGs) [13], [14], are introduced to describe the local characteristics of the SAR images. Recently, researchers have put forward the fusion of global and local features, which leads to superior effectiveness and robustness for SAR ATR [15].

Apart from these classic methods, another type of feature extraction scheme, namely dictionary learning (DL) and sparse representation (SR), has been demonstrated to be efficient in the community of signal representation and widely exploited for target recognition. In SR, the samples are approximated by a linear combination of a set of atoms with a sparsity constraint (ℓ_0- or ℓ_1-norm), such that the samples are mapped into a

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Lei Tao, Xue Jiang, and Xingzhao Liu are with the School of Electronic Information and Electrical Engineering, Shanghai Jiao Tong University, Shanghai 200240, China (e-mail: thunder_tao@sjtu.edu.cn; xuejiang@sjtu.edu.cn; xzhliu@sjtu.edu.cn).

Zhou Li is with the Beijing Institute of Remote Sensing Information, Beijing 100192, China (e-mail: zenithus@163.com).

Zhixin Zhou is with Space Engineering University, Beijing 101416, China (e-mail: zxzhou@cashq.ac.cn).

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sparse feature space. The dictionary, which is constituted on the basis of these redundant atoms, can be divided into linear and nonlinear ones. SR classification (SRC) [16] is a direct application of linear SR to a classification problem, whose dictionary is established using the original images [17]. However, the large dictionary of SRC contains redundant information and results in a heavy calculation burden. The K-singular value decomposition (SVD) algorithm [18] provides a solution to compute a more compact dictionary by performing SVD and each column of the dictionary is defined as the first column of the orthogonal matrix at each iteration, such that it can exploit a rank-one matrix to approximate the original image matrix. Based on this algorithm, the discriminative K-SVD (D-KSVD) [19] adds a classification error term into the objective function to improve the classification performance, while the label consistent K-SVD (LC-KSVD) [20] introduces a discriminative error term so that the features obtained from the same class have similar sparse codes. In addition, another structured DL, called Fisher discrimination DL (FDDL), has been proposed to design the dictionary atoms corresponding to the class labels [21]. In our previous work, we have proposed a multiscale incremental DL (MSI-DL) method [22], where a multidictionary model is learned in terms of an incremental supervised DL for ATR. However, as demonstrated in [23], the aforementioned linear representation methods are usually inadequate for representing nonlinear structure of data, which often occurs in many practical applications. Hence, the kernel K-SVD algorithm is proposed for nonlinear DL. Furthermore, such nonlinear DL includes the correlation constrained structured kernel K-SVD (CCSK-KSVD) [24], which combines the nonlinear reconstruction error and the discriminative error term with a correlation constraint.

Recently, DL and SR have been widely exploited for SAR ATR and numerous extensions have been proposed based on the abovementioned conventional feature extraction and DL methods, such as coupled DL [25], SAR-HOG-based DL [14], steerable-wavelet-frames-based SR [26], and two-stage fusion of multifeature low-rank representation (TFMFLRR) [27]. The SR of monogenic signal, which is a multidimensional generalization of the analytic signal, has also been applied to deal with SAR ATR [28]. In practice, different from optical imaging, the SAR platform is usually installed on a satellite or airplane, leading to difficulties in obtaining targets. Therefore, the size of the data set of SAR images is often limited. For the small training set size condition, many data augmentation strategies have been proposed, including the basic operations, such as translation, rotation and mirror, adding speckle noise [29], and multilevel reconstruction using attributed scattering center [30], especially when using deep learning methods. However, to our best knowledge, there is no effective and computationally efficient algorithm for nonlinear DL with data augmentation.

Inspired by the aforementioned works, we propose a multinonlinear dictionaries scheme for SAR ATR. In contrast to single DL, we first introduce multiscale features into the objective function and the reconstruction error terms are optimized simultaneously to obtain a multidictionary for features in multiscale. The multiscale features of the SAR image, which are defined as the filtered images, are obtained by utilizing the Frost filters with different parameters. From one aspect of view, extracting multiscale features for each image can be considered as a data augmentation process. In addition, images with various degrees of blurring after introducing Frost filters lose part of local and detailed information, however, global features of targets could gain more attention. Hence, applying various weights between different reconstruction error terms can better suppress the speckle noise while rebuilding the target. Second, the PCA is performed to reduce the dimension of each scale feature to save the computational cost of the algorithm. Finally, we address the nonlinear situation motivated by kernel K-SVD to each dimension-reduced scale feature, such that our goal is to learn multinonlinear dictionaries for multiscale features.

Classification task of DL can be accomplished in different ways which are as follows.

1) DL and classification are two separate steps. The sparse codes of training samples are obtained by DL, and then the classifier is learned using these sparse codes as the input signals [31].
2) Exploiting the samples from the same class to learn a subdictionary for each class and the predicted label is determined by computing the class of subdictionary that could produce the minimum reconstruction error [32], [33].
3) Introducing the classification model into the objective function to learn the dictionary and classifier simultaneously [34], [35].

In this article, we employ the third scheme and add a classification error term into the objective function to learn a classifier, which can also make the multidictionary more discriminative. Therefore, the objective function of the proposed scheme contains multinonlinear reconstruction error terms and a classification error term. The main contributions of this article are summarized as follows.

1) The multiscale features extracted by Frost filters are introduced. The PCA is performed to reduce the dimension of multiscale features followed by nonlinear mapping functions, whereas multinonlinear dictionaries are learned to improve the performance of SAR ATR with speckle noise and limited training set size. A linear classifier defined in the classification error term is introduced into the objective function and learned simultaneously.
2) A general conclusion about the discrimination performance is given with proof.
3) Two algorithms have been proposed for solving the resultant optimization problem, i.e., multiscale supervised kernel K-SVD (MSK-KSVD) and multiscale supervised incremental kernel DL (MSIK-DL). Note that MSIK-DL is developed based on the idea that we previously proposed in [22]. They both have superior recognition performance. From the aspect of running time due to the calculation of the inverse matrix at each iteration in MSIK-DL, it costs longer running time than MSK-KSVD. However, MSIK-DL just reads a small batch size of training samples at each iteration such that

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it consumes less memory resources than MSK-KSVD, which needs all training samples at each iteration.

The remainder of this article is organized as follows. In Section II, we briefly review two representative DL methods. The proposed DL scheme for SAR ATR is presented in Section III. Section IV describes two algorithms for the proposed optimization problem, followed by the classification scheme in Section V. The experimental results and analysis are shown in Section VI. Finally, we conclude this article in Section VII.

II. Related Works

In this section, two representative works, i.e., SR-based linear and nonlinear DL methods, are briefly reviewed. Hereafter, an image with a size of $p \times q$ pixels will be reshaped into a column vector whose dimension is m ($m = p \times q$).

A. LC-KSVD

Given a training set of SAR images $\boldsymbol{Y} = [\boldsymbol{y}_1, \boldsymbol{y}_2, \ldots, \boldsymbol{y}_N] \in \mathbb{R}^{m \times N}$, where $\boldsymbol{y}_i$ is an m-dimensional input image and N represents the training set size, the basic idea for SR and DL is to minimize the image reconstruction error with a sparsity constraint to learn a dictionary and its corresponding sparse code, which is defined as

$$\min_{\boldsymbol{D},\boldsymbol{X}} \|\boldsymbol{Y} - \boldsymbol{D}\boldsymbol{X}\|_F^2, \quad \text{s.t. } \|\boldsymbol{x}_i\|_0 \leqslant T, \ i = 1, 2, \ldots, N \tag{1}$$

where the vector $\boldsymbol{x}_i$ is the ith column of the sparse code matrix $\boldsymbol{X} \in \mathbb{R}^{J \times N}$. The matrix $\boldsymbol{D} = [\boldsymbol{d}_1, \boldsymbol{d}_2, \ldots, \boldsymbol{d}_J] \in \mathbb{R}^{m \times J}$ is the sought dictionary, where J is the number of atoms of the dictionary, and T is the sparsity parameter. Taking into consideration the discrimination performance of samples from different classes, LC-KSVD algorithm [20], including LC-KSVD1 and LC-KSVD2, utilizes a discriminative error term in the objective function. The objective function of LC-KSVD2 can be expressed as

$$\begin{aligned} \min_{\boldsymbol{D},\boldsymbol{A},\boldsymbol{W},\boldsymbol{X}} & \|\boldsymbol{Y} - \boldsymbol{D}\boldsymbol{X}\|_F^2 + \alpha\|\boldsymbol{Q} - \boldsymbol{A}\boldsymbol{X}\|_F^2 + \beta\|\boldsymbol{H} - \boldsymbol{W}\boldsymbol{X}\|_F^2 \\ & \text{s.t. } \|\boldsymbol{x}_i\|_0 \leqslant T, \quad i = 1, 2, \ldots, N \end{aligned} \tag{2}$$

whose objective function contains three error terms: 1) the reconstruction error term $\|\boldsymbol{Y} - \boldsymbol{D}\boldsymbol{X}\|_F^2$; 2) the discriminative error term $\|\boldsymbol{Q} - \boldsymbol{A}\boldsymbol{X}\|_F^2$; and 3) the classification error term $\|\boldsymbol{H} - \boldsymbol{W}\boldsymbol{X}\|_F^2$. In the discriminative error term, $\boldsymbol{Q} \in \mathbb{R}^{J \times N}$ is the discriminative sparse codes matrix generated in terms of label consistency, and $\boldsymbol{A} \in \mathbb{R}^{J \times J}$ is a transformation matrix that enforces $\boldsymbol{A}\boldsymbol{X}$ to approximate the discriminative sparse codes matrix $\boldsymbol{Q}$. Hence, the dictionary learned according to (2) becomes more discriminative and the sparse codes from different classes are more dissimilar, which is of benefit to classification. The third error term, i.e., the classification error term, introduced into the objective function, where $\boldsymbol{H} \in \mathbb{R}^{M \times N}$ is the label matrix, is to learn a linear classifier $\boldsymbol{W} \in \mathbb{R}^{M \times J}$. Thus, LC-KSVD2 learns the dictionary $\boldsymbol{D}$ and the classifier simultaneously. Here, M is the number of classes of SAR targets. Compared with LC-KSVD2, the objective function of LC-KSVD1 does not have the classification error term. For a classification task, the classifier, such as a linear classifier, is learned separately according to the sparse codes calculated using the LC-KSVD1 dictionary.

B. Kernel K-SVD

LC-KSVD is a linear DL method which assumes that the samples lie in a linear subspace. However, linear scheme is not applicable if the classes are not linearly separable. In this case, the samples should be mapped into a higher dimensional Hilbert space $\mathcal{F}$ with a nonlinear mapping function $\Phi : \mathbb{R}^m \to \mathcal{F} \subset \mathbb{R}^{\widetilde{m}}$, where $\widetilde{m}$ is often much larger than m. Then the nonlinear mapping result of $\boldsymbol{Y}$ is expressed as $\Phi(\boldsymbol{Y}) = [\Phi(\boldsymbol{y}_1), \Phi(\boldsymbol{y}_2), \ldots, \Phi(\boldsymbol{y}_N)] \in \mathbb{R}^{\widetilde{m} \times N}$. The kernel function, $\kappa(\cdot, \cdot) : \mathcal{F} \otimes \mathcal{F} \to \mathbb{R}^1$, is defined as the inner product in the Hilbert space, i.e., $\kappa(\boldsymbol{y}_i, \boldsymbol{y}_j) = \langle \Phi(\boldsymbol{y}_i), \Phi(\boldsymbol{y}_j) \rangle$, which satisfies the Mercer's condition [36]. For all images, the kernel matrix is $\mathcal{K}(\boldsymbol{Y}, \boldsymbol{Y}) = [\boldsymbol{\mathcal{K}}_{ij}] = [\kappa(\boldsymbol{y}_i, \boldsymbol{y}_j)] = \Phi(\boldsymbol{Y})^T \Phi(\boldsymbol{Y})$. By introducing the kernel function into DL, the objective function of a nonlinear DL called kernel K-SVD [23] is given by

$$\begin{aligned} \min_{\boldsymbol{D},\boldsymbol{X}} \|\Phi(\boldsymbol{Y}) - \boldsymbol{D}\boldsymbol{X}\|_F^2 \quad & \text{s.t. } \|\boldsymbol{x}_i\|_0 \leqslant T, \ i = 1, \ldots, N \\ & \|\boldsymbol{d}_j\|_2^2 = 1, \quad j = 1, \ldots, J. \end{aligned} \tag{3}$$

Since the dimension of the Hilbert space $\mathcal{F}$ could be infinite and each column $\boldsymbol{d}_j$ should lie in it, calculating the optimal dictionary $\boldsymbol{D}$ is a challenging problem. To simplify the optimization problem (3), one can utilize the following transformation:

$$\boldsymbol{D}^\star = \Phi(\boldsymbol{Y})\boldsymbol{A}. \tag{4}$$

Then the problem is converted into computing the optimal coefficient matrix $\boldsymbol{A} \in \mathbb{R}^{N \times J}$, which can be solved exploiting the kernel K-SVD and kernel OMP algorithm. In practice, the Gaussian kernel $\kappa(\boldsymbol{y}_i, \boldsymbol{y}_j) = \exp(-((\|\boldsymbol{y}_i - \boldsymbol{y}_j\|^2)/\delta))$ and the polynomial kernel $\kappa(\boldsymbol{y}_i, \boldsymbol{y}_j) = (\langle \boldsymbol{y}_i, \boldsymbol{y}_j \rangle + \rho)^\eta$ are often utilized as the kernel functions, where δ, ρ, and η are the given parameters. The classifier can be learned using the sparse codes of samples after the kernel SR stage.

III. Proposed Scheme

For a target recognition task, although DL and classifier design are two separate steps, joint optimizing both of them can often avoid suboptimal solution and achieve better performance [19], [34]. In this article, we learn both the nonlinear dictionary and classification model simultaneously, which are obtained by solving the following essential problem:

$$\begin{aligned} \min_{\boldsymbol{D},\boldsymbol{W},\boldsymbol{X}} & \|\Phi(\boldsymbol{Y}) - \boldsymbol{D}\boldsymbol{X}\|_F^2 + \beta \sum_{i=1}^{N} \mathcal{L}(h_i, f(\boldsymbol{x}_i, \boldsymbol{W})) \\ & \text{s.t. } \|\boldsymbol{x}_i\|_0 \leqslant T \text{ and } \|\boldsymbol{W}\|_F^2 \leqslant \xi, \quad i = 1, 2, \ldots, N \end{aligned} \tag{5}$$

where $f(\cdot)$ represents a classifier, $\mathcal{L}(\cdot)$ is a classification error function, h_i is the label of the ith sample, and $\boldsymbol{W}$ is a weight matrix on $\boldsymbol{x}_i$. The parameter β is used to balance the reconstruction error term and classification error term. The constraint term $\|\boldsymbol{W}\|_F^2 \leqslant \xi$ is exploited to avoid overfitting. Further discussions of the two error terms are stated in Sections III-A and III-B.

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A. Nonlinear Reconstruction Error

In practice, SAR images are usually affected by various environmental factors, such as speckle noise and clutter. It is a challenging task to retain the information related to the targets and attributes in SAR images. In addition, lack of SAR targets images leads to inadequate training or overfitting for a target recognition training process.

To alleviate the effects of these two issues, we adopt a smooth function $G(\cdot)$ with different parameters to an SAR image $\boldsymbol{y}$ to obtain its multiscale features and introduce these multiscale features into DL to employ several reconstruction error terms such that multidictionary can be learned. Denote the current window of the SAR image as $\boldsymbol{I}_c$, the size of which is $(2M_1+1)\times(2N_1+1)$, the centered pixel is (m,n), and the pixel set is $\mathcal{W}=\{(i,j)|m-M_1\le i\le m+M_1, n-N_1\le j\le n+N_1\}$. To suppress multiplicative speckle noise, the nonlinear Frost filter [37]

$$G_\zeta(i,j)=e^{-\zeta C^2\sqrt{(i-m)^2+(j-n)^2}} \tag{6}$$

is selected as the smooth function, where C is defined by the ratio of the local standard deviation to the local mean of the current window $\boldsymbol{I}_c$ and ζ is a positive parameter to determine the degree of blur of an SAR image. Then, the kth multiscale feature of an SAR image is defined as

$$\boldsymbol{O}_k(m,n)=\frac{1}{\sum_{i,j\in\mathcal{W}}G_{\zeta_k}(i,j)}\sum_{i,j\in\mathcal{W}}G_{\zeta_k}(i,j)\boldsymbol{I}(i,j) \tag{7}$$

where $\boldsymbol{I}$ and $\boldsymbol{O}_k$ represent the input and output images, respectively, and G_{ζ_k} is the Frost filter with a parameter ζ_k. It can be found that the smaller the ζ_k is, the more blurring the filtered image $\boldsymbol{O}_k$ is. By stacking all columns of the filtered image $\boldsymbol{O}_k$ into a vector $\mathcal{G}_k(\boldsymbol{y})$, we can obtain $\mathcal{G}_k(\boldsymbol{Y})=[\mathcal{G}_k(\boldsymbol{y}_1),\mathcal{G}_k(\boldsymbol{y}_2),\ldots,\mathcal{G}_k(\boldsymbol{y}_N)]$.

Since the size of $\mathcal{G}_k(\boldsymbol{Y})$ is m by N, which is computationally expensive, the PCA algorithm is performed to reduce the dimension of each scale feature from m to $\widehat{m}$ ($\widehat{m}\ll m$) to save the computational cost, which is denoted as $\mathcal{P}(\mathcal{G}_k(\boldsymbol{Y}))=[\mathcal{P}(\mathcal{G}_k(\boldsymbol{y}_1)),\mathcal{P}(\mathcal{G}_k(\boldsymbol{y}_2)),\ldots,\mathcal{P}(\mathcal{G}_k(\boldsymbol{y}_N))]\in\mathbb{R}^{\widehat{m}\times N}$, and we define

$$\boldsymbol{P}_k=\mathcal{P}(\mathcal{G}_k(\boldsymbol{Y})) \tag{8a}$$

$$\boldsymbol{p}_k^i=\mathcal{P}(\mathcal{G}_k(\boldsymbol{y}_i)). \tag{8b}$$

Taking into account the nonlinearity of each dimension-reduced scale feature, we utilize the nonlinear mapping function $\Phi(\cdot)$ to the feature of each scale and the nonlinear reconstruction error terms are expressed as

$$\sum_{k=1}^{K}\alpha_k\|\Phi(\boldsymbol{P}_k)-\boldsymbol{D}_k\boldsymbol{X}\|_F^2 \tag{9}$$

where K is the number of the reconstruction error terms and α_k is a weight assigned to each reconstruction error term to control the contribution of the corresponding terms. According to (4), each nonlinear dictionary $\boldsymbol{D}_k$ should lie in the Hilbert space $\mathcal{F}$ with the base vectors $\{\Phi(\boldsymbol{p}_k^1),\Phi(\boldsymbol{p}_k^2),\ldots,\Phi(\boldsymbol{p}_k^N)\}$ and $\boldsymbol{D}_k=\Phi(\boldsymbol{P}_k)\boldsymbol{A}_k$. Hence, we can rewrite the formula (9) as

$$\sum_{k=1}^{K}\alpha_k\|\Phi(\boldsymbol{P}_k)-\Phi(\boldsymbol{P}_k)\boldsymbol{A}_k\boldsymbol{X}\|_F^2. \tag{10}$$

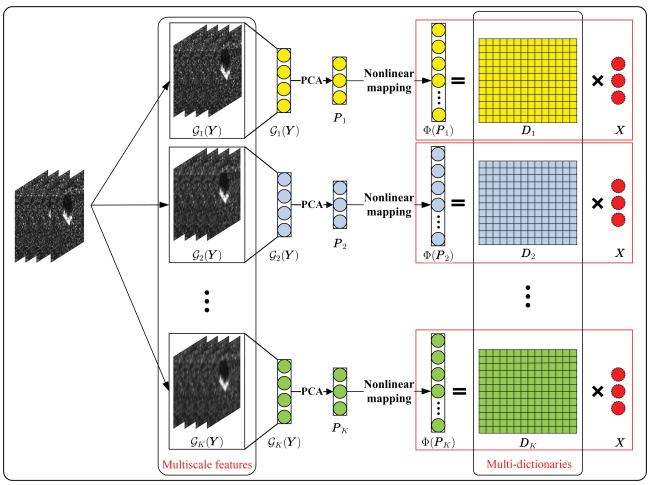


Fig. 1. Flowchart of multiple nonlinear reconstruction process. Nonlinear dimension-reduced multiscale features are reconstructed via the multidictionary $\{\boldsymbol{D}_1,\boldsymbol{D}_2,\ldots,\boldsymbol{D}_K\}$ and the sparse code matrix $\boldsymbol{X}$.

The flowchart of the proposed multiple nonlinear reconstruction process is shown in Fig. 1.

B. Discriminative Classification Error

The recognition performance mainly depends on the discrimination of the sparse codes. Besides the aforementioned nonlinear situation of SAR images in the reconstruction error terms, introducing a discriminative error term into the objective function is another strategy to improve the discrimination performance [19], [20], [24]. Then, the sparse codes from different classes are quite different and the samples with the same labels have similar sparse codes. Here, our goal is to combine the classification model with the discriminative error term to learn a discriminative classifier.

As stated in [38], the authors have proved that LC-KSVD is a reformulation of D-KSVD under identical initialization conditions. That is, LC-KSVD has a close discrimination performance as D-KSVD by adjusting the weights assigned to the error terms. However, these two methods are both single and linear DL. Next, we provide a more general conclusion about discrimination of nonlinear and multidictionary learning, which is given in Theorem 1.

Theorem 1: Assume that $\boldsymbol{Q}=[\boldsymbol{Q}_1,\boldsymbol{Q}_2,\ldots,\boldsymbol{Q}_M]\in\mathbb{R}^{J\times N}$ ($J\geqslant M$) is the discriminative matrix, where $\boldsymbol{Q}_i=[\boldsymbol{q}_{i,1},\boldsymbol{q}_{i,2},\ldots,\boldsymbol{q}_{i,N_i}]$ ($\sum_{i=1}^{M}N_i=N$) is the ith submatrix of $\boldsymbol{Q}$ and the corresponding samples are from class i. Each column of the discriminative matrix satisfies the following constraint:

$$\begin{cases}\langle\boldsymbol{q}_{i,p},\boldsymbol{q}_{j,q}\rangle=1, & \text{if } i=j\\ \langle\boldsymbol{q}_{i,p},\boldsymbol{q}_{j,q}\rangle=0, & \text{if } i\neq j.\end{cases} \tag{11}$$

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Then, for the following optimization problem:

$$\min_{\boldsymbol{D}_k,\boldsymbol{G},\boldsymbol{X}} \sum_{k=1}^{K} \alpha_k \|\Phi(\boldsymbol{P}_k) - \boldsymbol{D}_k \boldsymbol{X}\|_F^2 + \gamma \|\boldsymbol{Q} - \boldsymbol{G}\boldsymbol{X}\|_F^2 \quad \text{s.t. } \|\boldsymbol{x}_i\|_0 \leqslant T, \quad i = 1, 2, \ldots, N \tag{12}$$

the sought dictionaries $\{\boldsymbol{D}_k\}$ calculated using the algorithms in Section IV are the same no matter what the detailed form of $\boldsymbol{Q}$ is.

Proof: See the Appendix A.

Theorem 1 demonstrates that the discrimination performance will be the same as long as the discriminative code matrix satisfies the constraint (11). Hence, instead of designing a complicated form of the discriminative error term, we define the classification error term as

$$\sum_{i=1}^{N} \mathcal{L}(h_i, f(\boldsymbol{x}_i, \boldsymbol{W})) = \|\boldsymbol{H} - \boldsymbol{W}\boldsymbol{X}\|_F^2 \tag{13}$$

where $\boldsymbol{H} = [\boldsymbol{h}_1, \boldsymbol{h}_2, \ldots, \boldsymbol{h}_N]$ and $\boldsymbol{h}_i = [0, \ldots, 0, 1, 0, \ldots, 0]^T \in \mathbb{R}^M$ is an one-hot vector where the position of nonzero value represents the class. The label matrix also meets the constraint (11), such that the classification error term not only learns the classifier but also promotes the discrimination performance.

Therefore, the objective function of multiscale supervised kernel DL (MSK-DL) scheme combined with PCA is finally expressed as

$$\min_{\boldsymbol{A}_k,\boldsymbol{W},\boldsymbol{X}} \left\| \begin{bmatrix} \sqrt{\alpha_1}\Phi(\boldsymbol{P}_1) \\ \sqrt{\alpha_2}\Phi(\boldsymbol{P}_2) \\ \vdots \\ \sqrt{\alpha_K}\Phi(\boldsymbol{P}_K) \end{bmatrix} - \begin{bmatrix} \sqrt{\alpha_1}\Phi(\boldsymbol{P}_1)\boldsymbol{A}_1 \\ \sqrt{\alpha_2}\Phi(\boldsymbol{P}_2)\boldsymbol{A}_2 \\ \vdots \\ \sqrt{\alpha_K}\Phi(\boldsymbol{P}_K)\boldsymbol{A}_K \end{bmatrix} \boldsymbol{X} \right\|_F^2 + \beta \|\boldsymbol{H} - \boldsymbol{W}\boldsymbol{X}\|_F^2 \quad \text{s.t. } \|\boldsymbol{x}_i\|_0 \leqslant T \text{ and } \|\boldsymbol{W}\|_F^2 \leqslant \xi, \quad i = 1, 2, \ldots, N. \tag{14}$$

IV. Optimization Methods

In this section, we address two methods for solving the optimization problem (14). First, the reconstruction error term of (14) can be rewritten as

$$\left\| \begin{bmatrix} \sqrt{\alpha_1}\Phi(\boldsymbol{P}_1) \\ \vdots \\ \sqrt{\alpha_K}\Phi(\boldsymbol{P}_K) \end{bmatrix} - \begin{bmatrix} \sqrt{\alpha_1}\Phi(\boldsymbol{P}_1)\boldsymbol{A}_1 \\ \vdots \\ \sqrt{\alpha_K}\Phi(\boldsymbol{P}_K)\boldsymbol{A}_K \end{bmatrix} \boldsymbol{X} \right\|_F^2 = \left\| \begin{bmatrix} \sqrt{\alpha_1}\Phi(\boldsymbol{P}_1) \\ \vdots \\ \sqrt{\alpha_K}\Phi(\boldsymbol{P}_K) \end{bmatrix} - \begin{bmatrix} \Phi(\boldsymbol{P}_1) & & \\ & \ddots & \\ & & \Phi(\boldsymbol{P}_K) \end{bmatrix} \begin{bmatrix} \sqrt{\alpha_1}\boldsymbol{A}_1 \\ \vdots \\ \sqrt{\alpha_K}\boldsymbol{A}_K \end{bmatrix} \boldsymbol{X} \right\|_F^2. \tag{15}$$

Denote

$$\Psi(\boldsymbol{P}) = \left[\sqrt{\alpha_1}\Phi(\boldsymbol{P}_1), \sqrt{\alpha_2}\Phi(\boldsymbol{P}_2), \ldots, \sqrt{\alpha_K}\Phi(\boldsymbol{P}_K)\right]^T \tag{16a}$$

$$\Psi(\boldsymbol{p}^i) = \left[\sqrt{\alpha_1}\Phi(\boldsymbol{p}_1^i), \sqrt{\alpha_2}\Phi(\boldsymbol{p}_2^i), \ldots, \sqrt{\alpha_K}\Phi(\boldsymbol{p}_K^i)\right]^T \tag{16b}$$

$$\Gamma(\boldsymbol{P}) = \operatorname{diag}\{\Phi(\boldsymbol{P}_1), \Phi(\boldsymbol{P}_2), \ldots, \Phi(\boldsymbol{P}_{\mathrm{K}})\} \tag{16c}$$

$$\boldsymbol{B} = \left[\sqrt{\alpha_1}\boldsymbol{A}_1, \sqrt{\alpha_2}\boldsymbol{A}_2, \ldots, \sqrt{\alpha_K}\boldsymbol{A}_K\right]^T. \tag{16d}$$

Then, the objective function of (14) can be concisely written as

$$\min_{\boldsymbol{B},\boldsymbol{W},\boldsymbol{X}} \|\Psi(\boldsymbol{P}) - \Gamma(\boldsymbol{P})\boldsymbol{B}\boldsymbol{X}\|_F^2 + \beta\|\boldsymbol{H} - \boldsymbol{W}\boldsymbol{X}\|_F^2 \quad \text{s.t. } \|\boldsymbol{x}_i\|_0 \leqslant T \text{ and } \|\boldsymbol{W}\|_F^2 \leqslant \xi, \quad i = 1, 2, \ldots, N. \tag{17}$$

The training process of each method involves two stages: sparse coding and nonlinear dictionaries update.

A. Sparse Coding

The sparse coding refers to the calculation of the sparse code matrix $\boldsymbol{X}$ (or the sparse code vector $\boldsymbol{x}_i$) with other matrices and parameters fixed when solving (17). Note that the two Frobenius norm terms in the objective function of (17) can be combined together. Hence, we have

$$\boldsymbol{X} = \arg\min_{\boldsymbol{X}} \left\| \begin{bmatrix} \Psi(\boldsymbol{P}) \\ \sqrt{\beta}\boldsymbol{H} \end{bmatrix} - \begin{bmatrix} \Gamma(\boldsymbol{P}) & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{I} \end{bmatrix} \begin{bmatrix} \boldsymbol{B} \\ \sqrt{\beta}\boldsymbol{W} \end{bmatrix} \boldsymbol{X} \right\|_F^2 \quad \text{s.t. } \|\boldsymbol{x}_i\|_0 \leqslant T, \quad i = 1, 2, \ldots, N \tag{18}$$

which can be solved by the kernel orthogonal matching pursuit (KOMP) algorithm [23]. Denote

$$\widehat{\Psi}(\boldsymbol{P}) = \left[\Psi(\boldsymbol{P})^T, \sqrt{\beta}\boldsymbol{H}^T\right]^T \tag{19a}$$

$$\widehat{\Gamma}(\boldsymbol{P}) = \operatorname{diag}\{\Gamma(\boldsymbol{P}), \boldsymbol{I}\} \tag{19b}$$

$$\widehat{\boldsymbol{B}} = \left[\boldsymbol{B}^T, \sqrt{\beta}\boldsymbol{W}^T\right]^T \in \mathbb{R}^{(N\times K+M)\times J}. \tag{19c}$$

Notably, the kernel matrices exploited in the sparse coding stage are different from the initial matrices stated in [23]. These kernel matrices are

$$\widehat{\Psi}(\boldsymbol{P})^T\widehat{\Psi}(\boldsymbol{P}) = \Psi(\boldsymbol{P})^T\Psi(\boldsymbol{P}) + \beta\boldsymbol{H}^T\boldsymbol{H} \tag{20a}$$

$$\widehat{\Psi}(\boldsymbol{P})^T\widehat{\Gamma}(\boldsymbol{P}) = \left[\Psi(\boldsymbol{P})^T\Gamma(\boldsymbol{P}), \sqrt{\beta}\boldsymbol{H}^T\right] \tag{20b}$$

$$\widehat{\Gamma}(\boldsymbol{P})^T\widehat{\Gamma}(\boldsymbol{P}) = \operatorname{diag}\left\{\Gamma(\boldsymbol{P})^{\mathrm{T}}\Gamma(\boldsymbol{P}), \boldsymbol{I}\right\} \tag{20c}$$

where

$$\Psi(\boldsymbol{P})^T\Psi(\boldsymbol{P}) = \alpha_1\mathcal{K}(\boldsymbol{P}_1, \boldsymbol{P}_1) + \cdots + \alpha_K\mathcal{K}(\boldsymbol{P}_K, \boldsymbol{P}_K) \tag{21a}$$

$$\Gamma(\boldsymbol{P})^T\Gamma(\boldsymbol{P}) = \operatorname{diag}\{\mathcal{K}(\boldsymbol{P}_1, \boldsymbol{P}_1), \ldots, \mathcal{K}(\boldsymbol{P}_{\mathrm{K}}, \boldsymbol{P}_{\mathrm{K}})\} \tag{21b}$$

$$\Psi(\boldsymbol{P})^T\Gamma(\boldsymbol{P}) = \left[\sqrt{\alpha_1}\mathcal{K}(\boldsymbol{P}_1, \boldsymbol{P}_1), \ldots, \sqrt{\alpha_K}\mathcal{K}(\boldsymbol{P}_K, \boldsymbol{P}_K)\right]. \tag{21c}$$

B. MSK-KSVD

When the sparse code matrix $\boldsymbol{X}$ is fixed, we propose the MSK-KSVD algorithm based on the kernel K-SVD to solve the following optimization problem with the regularization constraint term $\|\boldsymbol{W}\|_F^2$ being dropped:

$$\widehat{\boldsymbol{B}} = \arg\min_{\widehat{\boldsymbol{B}}} \left\|\widehat{\Psi}(\boldsymbol{P}) - \widehat{\Gamma}(\boldsymbol{P})\widehat{\boldsymbol{B}}\boldsymbol{X}\right\|_F^2. \tag{22}$$

The dictionaries and the weight matrix are updated by optimizing $\widehat{\boldsymbol{B}}$ instead. Although the form of (22) is similar to (3) with the substitution of (4), each coefficient matrix $\boldsymbol{A}_k$ is associated with its corresponding scale feature $\Phi(\boldsymbol{P}_k)$ and is independent of other scale features. Hence, the kernel K-SVD cannot be applied directly. Inspired by Aharon *et al.* [18] and Nguyen *et al.* [23], the matrix $\widehat{\boldsymbol{B}}$ is updated column by column. Denote $\widehat{\boldsymbol{b}}_j$ and $\boldsymbol{x}^j$ as the jth column of $\widehat{\boldsymbol{B}}$ and jth

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row of $\boldsymbol{X}$, respectively. Then, the error term of (22) can be rewritten as

$$\begin{aligned}\left\|\widehat{\Psi}(\boldsymbol{P})-\widehat{\Gamma}(\boldsymbol{P})\widehat{\boldsymbol{B}}\boldsymbol{X}\right\|_F^2 &= \left\|\left(\widehat{\Psi}(\boldsymbol{P})-\widehat{\Gamma}(\boldsymbol{P})\sum_{j\neq n}\widehat{\boldsymbol{b}}_j\boldsymbol{x}^j\right)-\widehat{\Gamma}(\boldsymbol{P})\widehat{\boldsymbol{b}}_n\boldsymbol{x}^n\right\|_F^2 \\ &= \left\|\boldsymbol{E}-\widehat{\Gamma}(\boldsymbol{P})\widehat{\boldsymbol{b}}_n\boldsymbol{x}^n\right\|_F^2 \end{aligned} \tag{23}$$

where $\boldsymbol{E}=\widehat{\Psi}(\boldsymbol{P})-\widehat{\Gamma}(\boldsymbol{P})\sum_{j\neq n}\widehat{\boldsymbol{b}}_j\boldsymbol{x}^j$ indicates the error term with the nth column of $\widehat{\boldsymbol{B}}$ being removed. Assume that $\boldsymbol{E}$ is fixed, such that the minimization problem is equivalent to finding the optimal $\widehat{\boldsymbol{b}}_n$ and $\boldsymbol{x}^n$. Notably, $\boldsymbol{X}$ is a sparse matrix and the zero elements in $\boldsymbol{x}^n$ have no contribution on the minimization problem (23). Therefore, the error matrix $\boldsymbol{E}$ can be shrunken by discarding the columns corresponding to the zero elements in $\boldsymbol{x}^n$. Define $\omega_n=\{i\mid \boldsymbol{x}^n(i)\neq 0, 1\leq i\leq N\}$ as the indices of nonzeros in $\boldsymbol{x}^n$. Let $\boldsymbol{\Omega}_n\in\mathbb{R}^{N\times|\omega_n|}$ be a matrix with 1's on the position $(\omega_n(i),i)$ and 0's elsewhere. Then, right multiplication with $\boldsymbol{\Omega}_n$ can discard the zero elements in $\boldsymbol{x}^n$ and the corresponding columns in $\boldsymbol{E}$, which are further represented as $\boldsymbol{x}_R^n=\boldsymbol{x}^n\boldsymbol{\Omega}_n$ and $\boldsymbol{E}^R=\boldsymbol{E}\boldsymbol{\Omega}_n$. The optimization problem for updating the nth column of $\widehat{\boldsymbol{B}}$ can be expressed as

$$\widehat{\boldsymbol{b}}_n=\arg\min_{\widehat{\boldsymbol{b}}_n}\left\|\boldsymbol{E}^R-\widehat{\Gamma}(\boldsymbol{P})\widehat{\boldsymbol{b}}_n\boldsymbol{x}_R^n\right\|_F^2. \tag{24}$$

That is, we need to find a rank-one matrix $\widehat{\Gamma}(\boldsymbol{P})\widehat{\boldsymbol{b}}_n\boldsymbol{x}_R^n$ to approximate $\boldsymbol{E}^R$ in the sense of the Frobenius norm and the sought column $\widehat{\Gamma}(\boldsymbol{P})\widehat{\boldsymbol{b}}_n$ should be normalized in terms of ℓ_2-norm. According to the SVD decomposition, the reduced matrix $\boldsymbol{E}^R$ can be decomposed as

$$\boldsymbol{E}^R=\boldsymbol{U}\boldsymbol{\Sigma}\boldsymbol{V}^T=\sum_i\sigma_i\boldsymbol{u}_i\boldsymbol{v}_i^T \tag{25}$$

where $\boldsymbol{U}$ and $\boldsymbol{V}$ are two orthogonal matrices. Define $\boldsymbol{u}_1$ and $\boldsymbol{v}_1$ as the first column of $\boldsymbol{U}$ and $\boldsymbol{V}$ corresponding to the largest singular value $\sigma_1=\boldsymbol{\Sigma}(1,1)$. Then, the matrix $\sigma_1\boldsymbol{u}_1\boldsymbol{v}_1^T$ is the best rank-one approximation to $\boldsymbol{E}^R$, and we obtain

$$\widehat{\Gamma}(\boldsymbol{P})\widehat{\boldsymbol{b}}_n=\boldsymbol{u}_1 \tag{26a}$$

$$\boldsymbol{x}_R^n=\sigma_1\boldsymbol{v}_1^T. \tag{26b}$$

However, the error term $\boldsymbol{E}^R$ is an implicit matrix and cannot be decomposed directly using SVD. Hence, we compute the Gram matrix $(\boldsymbol{E}^R)^T\boldsymbol{E}^R=\boldsymbol{V}\boldsymbol{\Delta}\boldsymbol{V}^T=\boldsymbol{\Omega}_n^T\boldsymbol{E}^T\boldsymbol{E}\boldsymbol{\Omega}_n$, where

$$\begin{aligned}\boldsymbol{E}^T\boldsymbol{E}=\widehat{\Psi}(\boldsymbol{P})^T\widehat{\Psi}(\boldsymbol{P})&+\left(\sum_{j\neq n}\widehat{\boldsymbol{b}}_j\boldsymbol{x}^j\right)^T\widehat{\Gamma}(\boldsymbol{P})^T\widehat{\Gamma}(\boldsymbol{P})\sum_{j\neq n}\widehat{\boldsymbol{b}}_j\boldsymbol{x}^j \\ &-\left(\sum_{j\neq n}\widehat{\boldsymbol{b}}_j\boldsymbol{x}^j\right)^T\widehat{\Gamma}(\boldsymbol{P})^T\widehat{\Psi}(\boldsymbol{P})-\widehat{\Psi}(\boldsymbol{P})^T\widehat{\Gamma}(\boldsymbol{P})\sum_{j\neq n}\widehat{\boldsymbol{b}}_j\boldsymbol{x}^j\end{aligned} \tag{27}$$

and $\boldsymbol{\Delta}=\boldsymbol{\Sigma}^2$. With the substitution of (20) and (21) into (27) and performing eigenvalue decomposition (EVD) on $(\boldsymbol{E}^R)^T\boldsymbol{E}^R$, $\sigma_1=(\boldsymbol{\Delta}(1,1))^{1/2}$ and $\boldsymbol{v}_1$ are obtained. By multiplying $\boldsymbol{v}_1$ on both sides of (25), we obtain

$$\boldsymbol{E}^R\boldsymbol{v}_1=\sigma_1\boldsymbol{u}_1. \tag{28}$$

To calculate $\widehat{\boldsymbol{b}}_n$, we substitute (16) and (26a) into (28), which can be further expressed as

$$\begin{aligned}&\begin{bmatrix}\sqrt{\alpha_1}\Phi(\boldsymbol{P}_1)-\sqrt{\alpha_1}\Phi(\boldsymbol{P}_1)\sum_{j\neq n}(\boldsymbol{a}_1)_j\boldsymbol{x}^j\\ \vdots\\ \sqrt{\alpha_K}\Phi(\boldsymbol{P}_K)-\sqrt{\alpha_K}\Phi(\boldsymbol{P}_K)\sum_{j\neq n}(\boldsymbol{a}_K)_j\boldsymbol{x}^j\\ \sqrt{\beta}\boldsymbol{H}-\sqrt{\beta}\sum_{j\neq n}\boldsymbol{w}_j\boldsymbol{x}^j\end{bmatrix}\boldsymbol{\Omega}_n\boldsymbol{v}_1 \\ &\qquad=\sigma_1\begin{bmatrix}\sqrt{\alpha_1}\Phi(\boldsymbol{P}_1)(\boldsymbol{a}_1)_n\\ \vdots\\ \sqrt{\alpha_K}\Phi(\boldsymbol{P}_K)(\boldsymbol{a}_K)_n\\ \sqrt{\beta}\boldsymbol{w}_n\end{bmatrix}\end{aligned} \tag{29}$$

where $(\boldsymbol{a}_k)_j$ and $\boldsymbol{w}_j$ are the jth column of the coefficient matrix $\boldsymbol{A}_k$ and $\boldsymbol{W}$, respectively. Equation (29) leads to

$$\begin{cases}(\boldsymbol{a}_k)_n=\sigma_1^{-1}\left(\boldsymbol{I}-\sum_{j\neq n}(\boldsymbol{a}_k)_j\boldsymbol{x}^j\right)\boldsymbol{\Omega}_n\boldsymbol{v}_1\\ \boldsymbol{w}_n=\sigma_1^{-1}\left(\boldsymbol{H}-\sum_{j\neq n}\boldsymbol{w}_j\boldsymbol{x}^j\right)\boldsymbol{\Omega}_n\boldsymbol{v}_1.\end{cases} \tag{30}$$

It should be noted that the initial values of $\boldsymbol{B}^{(0)}$ (or $\boldsymbol{A}_k^{(0)}$) and $\boldsymbol{W}^{(0)}$ are vital for convergence. Here, we use a subset $\boldsymbol{Y}'$ of the training samples and randomly select an one-hot vector to initialize $\boldsymbol{A}_k^{(0)}$. The corresponding sparse code matrix $\boldsymbol{X}'$ is calculated using the KOMP algorithm by removing the classification error term in (18) and $\boldsymbol{W}^{(0)}$ can be calculated as follows [20]:

$$\boldsymbol{W}^{(0)}=\boldsymbol{H}'\boldsymbol{X}'^T\left(\boldsymbol{X}'\boldsymbol{X}'^T+\lambda\boldsymbol{I}\right)^{-1}. \tag{31}$$

The overall procedure of the proposed MSK-KSVD algorithm is summarized in Algorithm 1.

C. MSIK-DL

The proposed MSK-KSVD algorithm accesses all training samples at each iteration. Here, we provide a novel online method which reads a small batch of training samples at each iteration for dictionaries update. The optimization problem (17) can be rewritten as

$$\begin{aligned}&\min_{\boldsymbol{B},\boldsymbol{W}}\sum_{i=1}^N\mathcal{E}_i\{\boldsymbol{B},\boldsymbol{x}_i,\boldsymbol{h}_i,\boldsymbol{W}\}+\frac{\lambda}{2}\|\boldsymbol{W}\|_F^2 \\ &\text{s.t. }\boldsymbol{x}_i=\arg\min_{\boldsymbol{x}}\frac{1}{2}\|\Psi\left(\boldsymbol{p}^i\right)-\Gamma(\boldsymbol{P})\boldsymbol{B}\boldsymbol{x}\|_F^2+\gamma\|\boldsymbol{x}\|_0 \\ &\qquad i=1,2,\ldots,N.\end{aligned} \tag{32}$$

Here, we define $\mathcal{E}_i\{\boldsymbol{B},\boldsymbol{x}_i,\boldsymbol{h}_i,\boldsymbol{W}\}=(1/2)\|\boldsymbol{h}_i-\boldsymbol{W}\boldsymbol{x}_i\|_2^2$. Each column of $\Gamma(\boldsymbol{P})\boldsymbol{B}$ should be normalized in terms of the ℓ_2-norm to make the solution unique.

We consider a special case that only one sample is read at each iteration, such that the loss function of (32) of the ith sample is $\mathcal{L}_i\{\boldsymbol{B},\boldsymbol{x}_i,\boldsymbol{h}_i,\boldsymbol{W}\}=\mathcal{E}_i\{\boldsymbol{B},\boldsymbol{x}_i,\boldsymbol{h}_i,\boldsymbol{W}\}+(\lambda/2)\|\boldsymbol{W}\|_F^2$. According to [22], we need to compute the gradient of $\mathcal{L}_i$ with respect to $\boldsymbol{B}$ and $\boldsymbol{W}$ and utilize the stochastic gradient descent scheme to update them. The chain rule $(\partial\mathcal{L}_i/\partial\boldsymbol{B}_{pq})=(\partial\mathcal{L}_i/\partial\boldsymbol{x}_i^T)(\partial\boldsymbol{x}_i/\partial\boldsymbol{B}_{pq})$ is used to establish bridges between

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Algorithm 1 MSK-KSVD

Input: SAR training samples $\boldsymbol{Y}$, label matrix $\boldsymbol{H}$, number of multiscale features K, number of training samples N, parameters J, T, $\{\alpha_k\}$, β, number of iteration L

Initialize: Compute the multiscale features of $\boldsymbol{Y}$ and perform dimension reduction to obtain $\{\boldsymbol{P}_k\}$. Utilize one-hot vector to initialize each column of $\{\boldsymbol{A}_k^{(0)}\}$, and $\boldsymbol{W}^{(0)}$ is obtained according to (31). Compute the kernel matrices $\{\mathcal{K}(\boldsymbol{P}_k, \boldsymbol{P}_k)\}$. Set iteration $l = 1$.
for $l \leqslant L$ **do**
  Set $n = 1$;
  Step 1: Sparse coding
  Utilize the KOMP algorithm introduced in Section IV-A to compute $\boldsymbol{X}^{(l)}$ with fixed $\boldsymbol{A}_k^{(l-1)}$ and $\boldsymbol{W}^{(l-1)}$.
  Step 2: Dictionaries and weight matrix update
  Define $(\boldsymbol{a}_k)_n^{(l)}$ and $\boldsymbol{w}_n^{(l)}$ as the nth column of $\boldsymbol{A}_k^{(l)}$ and $\boldsymbol{W}^{(l)}$, respectively.
  for $n \leqslant J$ **do**
    Find the indices w_n of nonzeros in $(\boldsymbol{x}^n)^{(l)}$ and transform it into the matrix $\boldsymbol{\Omega}_n$.
    Compute the Gram matrix $(\boldsymbol{E}^R)^T\boldsymbol{E}^R$ according to (27) and $\boldsymbol{\Omega}_n$.
    Perform EVD on $(\boldsymbol{E}^R)^T\boldsymbol{E}^R$ to obtain $\boldsymbol{v}_1$ and σ_1.
    Update $(\boldsymbol{a}_k)_n^{(l)}$ and $\boldsymbol{w}_n^{(l)}$ according to (30).
    $n = n + 1$;
  end
  $l = l + 1$;
end

Output: Trained coefficient matrices $\{\boldsymbol{A}_k\}$ and weight matrix $\boldsymbol{W}$

$\mathcal{L}_i$ and $\boldsymbol{B}$, where $(\partial\mathcal{L}_i/\partial\boldsymbol{x}_i) = \boldsymbol{W}^T(\boldsymbol{W}\boldsymbol{x}_i - \boldsymbol{h}_i)$ and $\boldsymbol{B}_{pq}$ represents the element in the position (p, q) of the matrix $\boldsymbol{B}$. Next, we consider how to compute $(\partial\boldsymbol{x}_i/\partial\boldsymbol{B}_{pq})$. Using $\|\boldsymbol{x}_i\|_1$ to replace $\|\boldsymbol{x}_i\|_0$, we define $E(\boldsymbol{B}, \boldsymbol{x}_i) = (1/2)\|\Psi(\boldsymbol{p}^i) - \Gamma(\boldsymbol{P})\boldsymbol{B}\boldsymbol{x}_i\|_F^2 + \gamma\|\boldsymbol{x}_i\|_1$. An implicit differentiation method is used to establish connection between $\boldsymbol{x}_i$ and $\boldsymbol{B}$. Assuming that $\boldsymbol{B}$ is fixed, take the derivative of $E(\boldsymbol{B}, \boldsymbol{x}_i)$ with respect to $\boldsymbol{x}_i$ and when it achieves its minimum value, we can obtain $\boldsymbol{B}^T\Gamma(\boldsymbol{P})^T(\Gamma(\boldsymbol{P})\boldsymbol{B}\boldsymbol{x}_i - \Psi(\boldsymbol{p}^i)) = -\gamma\,\mathrm{sgn}(\boldsymbol{x}_i)$. Then, take the derivative with respect to $\boldsymbol{B}_{pq}$ on both sides, we have

$$\frac{\partial\left[\boldsymbol{B}^T\Gamma(\boldsymbol{P})^T\left(\Gamma(\boldsymbol{P})\boldsymbol{B}\boldsymbol{x}_i - \Psi\left(\boldsymbol{p}^i\right)\right)\right]}{\partial\boldsymbol{B}_{pq}} = \frac{\partial\left[-\gamma\,\mathrm{sgn}(\boldsymbol{x}_i)\right]}{\partial\boldsymbol{B}_{pq}}. \quad (33)$$

The term on the left side of (33) cannot be infinite, therefore, $\partial[\boldsymbol{B}^T\Gamma(\boldsymbol{P})^T(\Gamma(\boldsymbol{P})\boldsymbol{B}\boldsymbol{x}_i - \Psi(\boldsymbol{p}^i))]/\partial\boldsymbol{B}_{pq} = 0$. Let Λ be the index set of nonzero values of $\boldsymbol{x}_i$ and $\bar{\Lambda}$ be the index set of zero values of $\boldsymbol{x}_i$. Define another auxiliary variable $\boldsymbol{\varphi} \in \mathbb{R}^J$, where $\boldsymbol{\varphi}_\Lambda = ((\boldsymbol{B}_\Lambda)^T\Gamma(\boldsymbol{P})^T\Gamma(\boldsymbol{P})\boldsymbol{B}_\Lambda)^{-1}(\partial\mathcal{L}_i/\partial(\boldsymbol{x}_i)_\Lambda)$ and $\boldsymbol{\varphi}_{\bar{\Lambda}} = 0$. Here, the kernel matrix $\Gamma(\boldsymbol{P})^T\Gamma(\boldsymbol{P})$ is defined in (20) and $\boldsymbol{B}_\Lambda$ is obtained by selecting the columns of $\boldsymbol{B}$ with index Λ.[1]

[1]For example, if $\boldsymbol{x}_i = [1, 1, 1, 0, 1, 0, 0, 1]^T$ and $\boldsymbol{B} = [\boldsymbol{b}_1, \boldsymbol{b}_2, \ldots, \boldsymbol{b}_8]$, the index set of zero values of $\boldsymbol{x}_i$ is $\Lambda = \{1, 2, 3, 5, 8\}$ and the corresponding matrix $\boldsymbol{B}_\Lambda = [\boldsymbol{b}_1, \boldsymbol{b}_2, \boldsymbol{b}_3, \boldsymbol{b}_5, \boldsymbol{b}_8]$.

Thus, the gradients of $\mathcal{L}_i$ with respect to $\boldsymbol{B}$ and $\boldsymbol{W}$ for the ith sample are

$$\frac{\partial\mathcal{L}_i}{\partial\boldsymbol{B}} = -\Gamma(\boldsymbol{P})^T\Gamma(\boldsymbol{P})\boldsymbol{B}\boldsymbol{\varphi}\boldsymbol{x}_i^T + \left[\Gamma(\boldsymbol{P})^T\Psi\left(\boldsymbol{p}^i\right) - \Gamma(\boldsymbol{P})^T\Gamma(\boldsymbol{P})\boldsymbol{B}\boldsymbol{x}_i\right]\boldsymbol{\varphi}^T \quad (34a)$$

$$\frac{\partial\mathcal{L}_i}{\partial\boldsymbol{W}} = (\boldsymbol{W}\boldsymbol{x}_i - \boldsymbol{h}_i)\boldsymbol{x}_i^T + \lambda\boldsymbol{W}. \quad (34b)$$

The details of the solution procedure are derived in the Appendix B. Two vital points should be noted which are as follows.

1) Different from the sparse coding stage used in the MSK-KSVD, the objective function of sparse coding here does not contain the classification error term, which is represented as

$$\boldsymbol{x} = \arg\min_{\boldsymbol{x}} \frac{1}{2}\|\Psi(\boldsymbol{p}) - \Gamma(\boldsymbol{P})\boldsymbol{B}\boldsymbol{x}\|_F^2, \quad \text{s.t. } \|\boldsymbol{x}\|_0 \leqslant T. \quad (35)$$

The optimization problem (35) can be solved utilizing the KOMP algorithm with the kernel matrices listed in (21).

2) When a small batch of training samples $\boldsymbol{Y}'' \in \mathbb{R}^{m\times N_1}$, i.e., $N_1 < N$, are considered at each iteration, a similar procedure can be derived. We just need to replace $\boldsymbol{h}_i$, $\boldsymbol{x}_i$, and $\boldsymbol{y}_i$ with $\boldsymbol{H}''$, $\boldsymbol{X}''$, and $\boldsymbol{Y}''$ to obtain the gradient $(\partial\mathcal{L}_i/\partial\boldsymbol{B})$ and $(\partial\mathcal{L}_i/\partial\boldsymbol{W})$, where $\boldsymbol{H}''$ and $\boldsymbol{X}''$ are the corresponding label matrix and sparse code matrix of $\boldsymbol{Y}''$.

To conclude, the matrices $\boldsymbol{B}$ and $\boldsymbol{W}$ are updated incrementally using online learning which reads a small batch of training samples at each iteration. The initializations of $\boldsymbol{B}^{(0)}$ (or $\boldsymbol{A}_k^{(0)}$) and $\boldsymbol{W}^{(0)}$ are the same as the aforementioned MSK-KSVD algorithm. We use the stochastic gradient descent algorithm to update $\boldsymbol{B}$ and $\boldsymbol{W}$ in terms of the results of (34). The learning rate τ is set to a small value and the columns of the dictionary $\Gamma(\boldsymbol{P})\boldsymbol{B}$ are normalized on the basis of ℓ_2-norm at each iteration. The whole process of the proposed MSIK-DL is summarized in Algorithm 2.

V. Classification Scheme

During the training stage, the optimization problem (17) considers the nonlinear reconstruction error terms and the discriminative classification error term, simultaneously. The nonlinear dictionaries $\boldsymbol{D}_k = \Phi(\boldsymbol{P}_k)\boldsymbol{A}_k$ and the weight matrix $\boldsymbol{W}$ are learned using the proposed MSK-KSVD or MSIK-DL algorithm. In the testing stage, given a testing SAR image $\boldsymbol{y}$ without label information, only nonlinear reconstruction error terms are taken into account to compute its SR $\boldsymbol{x}$.

Notably, in the MSK-KSVD algorithm, the sought matrices $\{\boldsymbol{A}_k\}$ and $\boldsymbol{W}$ are treated as a whole matrix $\widehat{\boldsymbol{B}}$, and each column of $\widehat{\Gamma}(\boldsymbol{P})\widehat{\boldsymbol{B}}$ is normalized according to

$$\left\|\begin{bmatrix}\Gamma(\boldsymbol{P})\boldsymbol{b}_j \\ \boldsymbol{w}_j\end{bmatrix}\right\|_2 = 1. \quad (36)$$

However, in the testing stage, when solving the optimization problem (35) to obtain the sparse code $\boldsymbol{x}$, the columns of the

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Algorithm 2 MSIK-DL (One Sample at Each Iteration)

Input: $\boldsymbol{Y}$, $\boldsymbol{H}$, K, N, parameters J, T, $\{\alpha_k\}$, λ, γ, τ, number of iteration L

Initialize: Compute the multiscale features of $\boldsymbol{Y}$ and perform dimension reduction to obtain $\{\boldsymbol{P}_k\}$. Utilize one-hot vector to initialize each column of $\{\boldsymbol{A}_k^{(0)}\}$ and $\boldsymbol{W}^{(0)}$ is obtained according to (31). Compute the kernel matrices $\{\mathcal{K}(\boldsymbol{P}_k, \boldsymbol{P}_k)\}$. Set $l = 0$.
for $l \leqslant L$ **do**
  Set $i = 1$;
  for $i \leqslant N$ **do**
    Compute the kernel matrices $\{\mathcal{K}(\boldsymbol{p}_k^i, \boldsymbol{P}_k)\}$.
    Compute the sparse code $\boldsymbol{x}_i$ using KOMP with new kernel matrices in (21).
    Employ (34) to calculate the gradient of $\mathcal{L}_i$ with respect to $\boldsymbol{B}^{(l)}$ and $\boldsymbol{W}^{(l)}$, respectively.
    Update $\boldsymbol{B}^{(l)} = \boldsymbol{B}^{(l)} - \tau \frac{\partial \mathcal{L}_i}{\partial \boldsymbol{B}^{(l)}}$;
    Update $\boldsymbol{W}^{(l)} = \boldsymbol{W}^{(l)} - \tau \frac{\partial \mathcal{L}_i}{\partial \boldsymbol{W}^{(l)}}$.
    Normalize each column of $\Gamma(\boldsymbol{P})\boldsymbol{B}^{(l)}$ in terms of the ℓ_2-norm by updating each column of $\boldsymbol{B}^{(l)}$, which can be computed by $\boldsymbol{b}_j^{(l)} = \frac{\boldsymbol{b}_j^{(l)}}{\sqrt{(\boldsymbol{b}_j^{(l)})^T \Gamma(\boldsymbol{P})^T \Gamma(\boldsymbol{P}) \boldsymbol{b}_j^{(l)}}}$.
  end
  Update $\boldsymbol{B}^{(l+1)} = \boldsymbol{B}^{(l)}$ and $\boldsymbol{W}^{(l+1)} = \boldsymbol{W}^{(l)}$;
  $l = l + 1$;
end

Output: Trained coefficient matrices $\{\boldsymbol{A}_k\}$ and weight matrix $\boldsymbol{W}$

learned dictionary $\Gamma(\boldsymbol{P})\boldsymbol{B}$ should satisfy the unit-norm condition [23]. Hence, each column of $\boldsymbol{b}_j$ should be normalized in terms of ℓ_2-norm as follows:

$$\boldsymbol{b}_j^\star = \frac{\boldsymbol{b}_j}{\sqrt{\boldsymbol{b}_j^T \Gamma(\boldsymbol{P})^T \Gamma(\boldsymbol{P}) \boldsymbol{b}_j}} \tag{37}$$

leading to

$$\boldsymbol{w}_j^\star = \frac{\boldsymbol{w}_j}{\sqrt{\boldsymbol{b}_j^T \Gamma(\boldsymbol{P})^T \Gamma(\boldsymbol{P}) \boldsymbol{b}_j}}. \tag{38}$$

If the MSIK-DL algorithm is used to train these dictionaries, we do not need to perform (37) and (38) since each column of $\Gamma(\boldsymbol{P})\boldsymbol{B}$ has been normalized at each iteration, resulting in $\boldsymbol{b}_j^\star = \boldsymbol{b}_j$ and $\boldsymbol{w}_j^\star = \boldsymbol{w}_j$.

Thus, in the testing stage, we first calculate the given testing sample's multiscale features $\mathcal{G}_k(\boldsymbol{y})$ and use PCA to reduce the dimension of each scale feature as $\boldsymbol{p}_k = \mathcal{P}(\mathcal{G}_k(\boldsymbol{y}))$. Then, the SR $\boldsymbol{x}$ of $\boldsymbol{y}$ is computed with the dictionary $\Gamma(\boldsymbol{Y})\boldsymbol{B}^\star$ by solving the following optimization problem:

$$\boldsymbol{x} = \arg\min_{\boldsymbol{x}} \frac{1}{2} \|\Psi(\boldsymbol{p}) - \Gamma(\boldsymbol{P})\boldsymbol{B}^\star \boldsymbol{x}\|_F^2, \quad \text{s.t. } \|\boldsymbol{x}\|_0 \leqslant T. \tag{39}$$

Finally, the predicted label vector of $\boldsymbol{y}$ is $\widehat{\boldsymbol{h}} = \boldsymbol{W}^\star \boldsymbol{x}$ and the label is the index of maximum value of $\widehat{\boldsymbol{h}}$. The whole process of the proposed methods for SAR target recognition is summarized in Algorithm 3.

Algorithm 3 MSK-DL for SAR Target Recognition

Input: $\boldsymbol{Y}$, $\boldsymbol{H}$, K, N, parameters J, T, $\{\alpha_k\}$, λ, γ, number of iteration L, SAR testing samples $\boldsymbol{Y}_{\text{test}}$

Initialize: Compute the multiscale features of $\boldsymbol{Y}$ and perform dimension reduction to obtain $\{\boldsymbol{P}_k\}$. Compute the kernel matrices $\{\mathcal{K}(\boldsymbol{P}_k, \boldsymbol{P}_k)\}$. Utilize one-hot vector to initialize each column of $\{\boldsymbol{A}_k^{(0)}\}$ and $\boldsymbol{W}^{(0)}$ is obtained according to (31). Compute the multiscale features of $\boldsymbol{Y}_{\text{test}}$ and perform PCA to obtain $\{(\boldsymbol{P}_{\text{test}})_k\}$.
Training Stage:
Scheme one (MSK-KSVD): Update $\{\boldsymbol{A}_k\}$ and $\boldsymbol{W}$ according to (30). Normalize each column of $\boldsymbol{B}$ as (37) and $\boldsymbol{W}$ is recalculated according to (38).
Scheme two (MSIK-DL): Update $\boldsymbol{B}$ and $\boldsymbol{W}$ incrementally according to (34).
Testing Stage:
Step 1: Calculate the kernel matrices $\Gamma(\boldsymbol{P})^T \Psi(\boldsymbol{P}_{\text{test}})$ and $\Psi(\boldsymbol{P}_{\text{test}})^T \Psi(\boldsymbol{P}_{\text{test}})$ according to (21). Compute the sparse code matrix $\boldsymbol{X}_{\text{test}}$ by using KOMP to solve (39).
Step 2: Predict the label matrix $\widehat{\boldsymbol{H}} = \boldsymbol{W}\boldsymbol{X}_{\text{test}}$.

Output: The predicted label matrix $\widehat{\boldsymbol{H}}$ of testing samples.

VI. Experimental Results

We evaluate the performance of the proposed scheme using the moving and stationary target automatic recognition (MSTAR) data set. To the best of our knowledge, it is the only publicly released and widely accepted data set available for SAR target recognition. This data set is a gallery collected SAR sensor using the X-band HH polarization mode with one foot resolution in range and azimuth. The images are captured at various aspect angles over 0°–360°. The images of one target are also obtained at different depression angles, such as 15°, 17°, and 30°. Each image in our experiments is cropped to 128 × 128 pixels in size. Fig. 2 shows some examples of the MSTAR targets at different aspect angles and depression angles. One usage of the MSTAR data set is under standard operating conditions (SOCs), where the number of aspect views for each target is listed in Table I. The samples acquired at depression angle 17° and 15° are utilized for training and testing, respectively. The entries in parentheses, such as SN_132 and SN_9563, represent the series of targets with various variations in configuration. In what follows, we will verify the convergence of the two proposed algorithms, evaluate the effect of relevant parameters on recognition performance, show the discrimination performance of different DL methods, and compare the proposed scheme with some representative methods. To fine-tune the relevant free parameters, we divide the samples at depression angle 17° into training and validation data sets, which are shown in Table II.

A. Convergence of the Two Proposed Algorithms

First, we demonstrate the convergence of the proposed MSK-KSVD and MSIK-DL algorithms. Using the samples shown in Table II, we record the values of the loss function on the training data and evaluate the recognition performance using the learned dictionaries on the validation data

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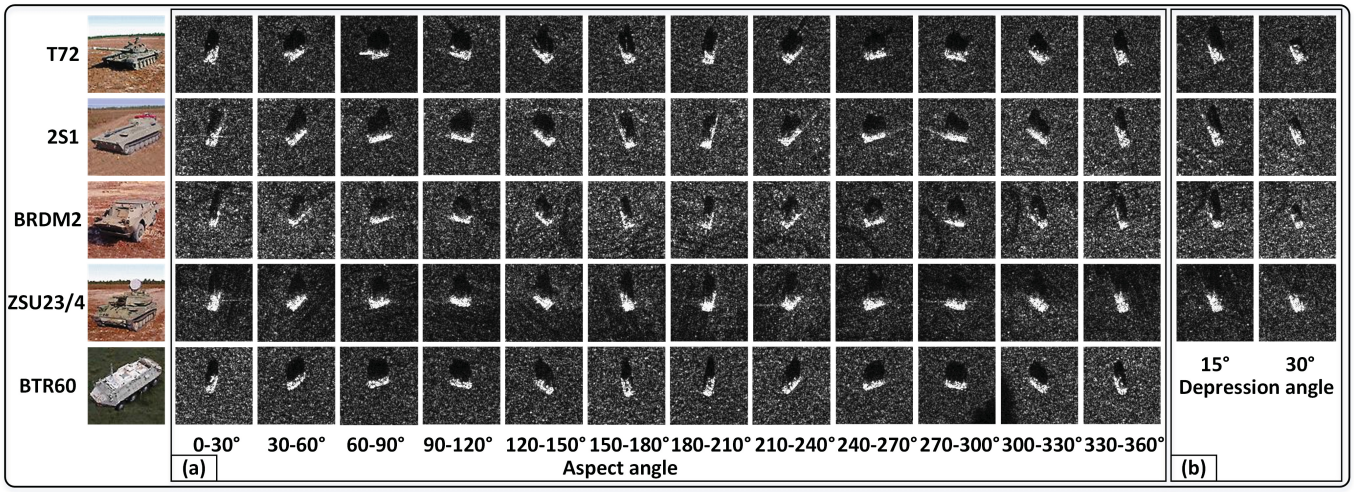


Fig. 2. Example images of MSTAR data set with different aspect angles and elevation angles. (a) Various aspect angles over 0°–360° with step size 30°. (b) Depression angles 15° and 30°.

TABLE I
NUMBER OF TARGETS OF TEN CLASSES AT DEPRESSION ANGLES 15° AND 17° UNDER SOC

Target		Train (17°)	Test (15°)	Target		Train (17°)	Test (15°)
TG1	T72	232 (SN_132)	196 (SN_132)	TG4	2S1	299	274
			195 (SN_812)	TG5	BRDM2	298	274
			191 (SN_S7)	TG6	BTR60	256	195
TG2	BMP2	233 (SN_9563)	195 (SN_9563)	TG7	D7	299	274
			196 (SN_9566)	TG8	T62	299	273
			196 (SN_C21)	TG9	ZIL131	299	274
TG3	BTR70	233	196	TG10	ZSU23/4	299	274

TABLE II
NUMBER OF TARGETS AT DEPRESSION ANGLE 17° FOR TRAINING AND VALIDATION

	TG1	TG2	TG3	TG4	TG5	TG6	TG7	TG8	TG9	TG10
Training	30	30	30	30	30	30	30	30	30	30
Validation	202	203	203	269	268	226	269	269	269	269

at each iteration. The polynomial function is empirically selected as the kernel function with $\rho = 0.2$ and $\eta = 4$. We utilize three multiscale features for each image and the square roots of weights on the three multicale features are $(\sqrt{\alpha_1}, \sqrt{\alpha_2}, \sqrt{\alpha_3}) = (1, 1, 1)$. The dimension of multiscale features is reduced to 20 by PCA operation. We set the dictionary size to 200. The sparsity factor T is set to 15. The weight β on the classification error term of the MSK-KSVD is 1. Instead of calculating the value of the loss function that contains an implicitly defined mapping in (22), the loss function of the proposed MSK-KSVD algorithm can be rewritten as

$$\begin{aligned}&\left\|\widehat{\Psi}(\boldsymbol{P}) - \widehat{\Gamma}(\boldsymbol{P})\widehat{\boldsymbol{B}}\boldsymbol{X}\right\|_F^2 \\ &= \mathrm{tr}\Big(\widehat{\Psi}(\boldsymbol{P})^T\widehat{\Psi}(\boldsymbol{P}) - \widehat{\Psi}(\boldsymbol{P})^T\widehat{\Gamma}(\boldsymbol{P})\boldsymbol{B}\boldsymbol{X} \\ &\quad - \boldsymbol{X}^T\boldsymbol{B}^T\widehat{\Gamma}(\boldsymbol{P})^T\widehat{\Psi}(\boldsymbol{P}) + \boldsymbol{X}^T\boldsymbol{B}^T\widehat{\Gamma}(\boldsymbol{P})^T\widehat{\Gamma}(\boldsymbol{P})\boldsymbol{B}\boldsymbol{X}\Big) \quad (40)\end{aligned}$$

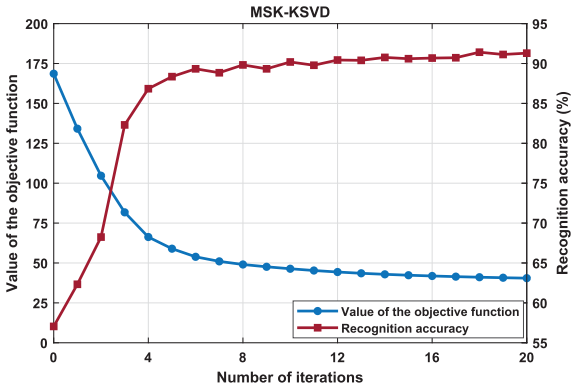


Fig. 3. Example of convergence of MSK-KSVD.

where tr(·) is the trace of a matrix. As a result, we use the kernel function for computing the value of the objective function. Fig. 3 shows the optimization process with the increase of iteration and exhibits a superior performance on convergence. For the MSIK-DL algorithm, the objective function is $(1/2)\|\boldsymbol{H} - \boldsymbol{W}\boldsymbol{X}\|_F^2 + (\lambda/2)\|\boldsymbol{W}\|_F^2$. The parameter λ of the regularization term is set to 10^{-6}. The learning rate τ has a great influence on the convergence speed. In order to find a reasonable learning rate, we consider three learning rates ($\tau = \{0.01, 0.05, 0.1\}$), and verify the impact on convergence and recognition accuracy, respectively, the results of which are shown in Fig. 4. We can observe that the value of the objective function decreases with the number of iterations increasing in all these three cases. Nevertheless, the convergence curve exhibits a local oscillation when $\tau = 0.1$. A too small learning rate, such as $\tau = 0.01$, leads to low convergence speed. Considering these two factors, we take $\tau = 0.05$ as a suitable option.

B. Effect of Relevant Parameters on Recognition

There are several relevant free parameters that affect the recognition performance. We perform the following

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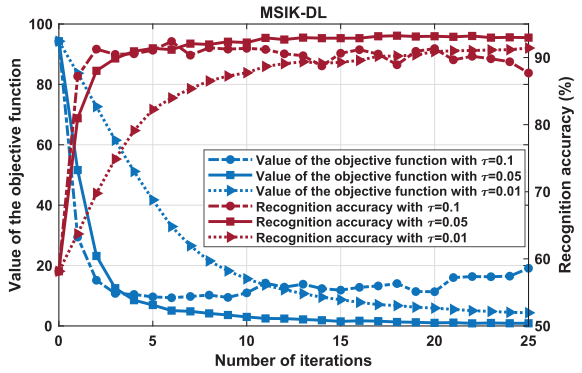


Fig. 4. Example of convergence of MSIK-DL.

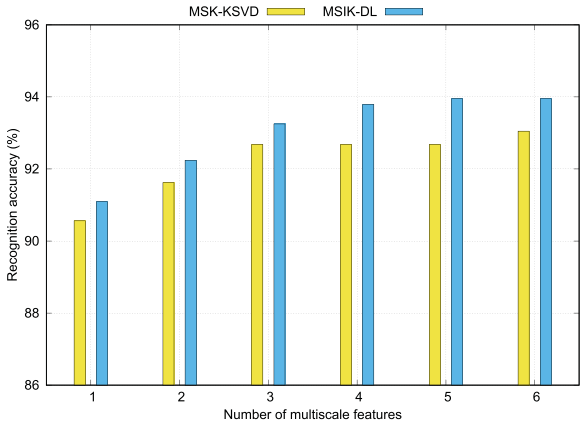


Fig. 5. Recognition accuracy versus number of multiscale feature with different numbers of training samples using MSK-KSVD and MSIK-DL.

TABLE III
SAR TARGET RECOGNITION ACCURACY USING THE MSK-KSVD AND MSIK-DL ALGORITHMS WITH VARYING DIMENSIONS OF EACH SCALE FEATURE

Dimension	10	15	20	25	30	35	40	45
MSK-KSVD	86.51	92.19	92.68	91.58	92.28	90.76	90.23	88.19
MSIK-DL	86.43	93.01	93.25	92.52	93.09	92.60	92.20	89.82

experiments using the training and validation data sets which are shown in Table II to evaluate and fine tune these parameters.

1) Dimension of Each Scale Feature: The first parameter that we evaluate is the dimension of each scale feature after performing the PCA algorithm, which is set from 10 to 45 for the two proposed algorithms. Other parameters are set as the above experimental settings. Table III compares the recognition results obtained using varying dimensions of each scale feature. It can be found that there is a significant difference with varying dimensions. The recognition accuracy first increases to a peak point and then decreases when further increasing the dimension of each scale feature, where the best recognition accuracies (92.68% and 93.25%) are obtained at the dimension being 20 for both the MSK-KSVD and MSIK-DL algorithms. The results suggest that the dimension is set to 20 for the forthcoming experiments. In addition, the recognition accuracies fluctuate slightly when the dimensions are between 15 and 30, which illustrates that the dimensions set within the range of 15–30 produce relatively high recognition results.

2) Number of Multiscale Features: In this part, we evaluate the influence of multiscale feature numbers on the recognition accuracy using MSK-KSVD and MSIK-DL. The dictionary size is set to 200. Different scale features are extracted by Frost filters with different parameters. The values of the parameter ζ_k are set to $\{\zeta_k\}_{k=1,2,\ldots,6} = \{4, 3, 2, 1, 0.5, 0.1\}$. The weight on each scale feature is set to 1. The other parameters are set as the previous settings. Fig. 5 exhibits the classification results using the two proposed algorithms. The single most striking observation to emerge from the bar chart comparison is that there is a remarkable improvement when increasing the number of multiscale feature from one to three. And the recognition accuracies tend to be stable as the multiscale feature number continues increasing. In addition, the recognition accuracies of the MSK-KSVD and MSIK-DL algorithms are close with a difference of about 1%, which can be explained by the fact that they are for solving the same optimization problem.

3) Weights: This set of analysis examines the impact of weights on recognition accuracies. Since a large number of multiscale features will lead to high computational cost and the time complexity is $O(z^{K-1})$, where z is the number of values that each weight is chosen from, it is nontrivial to find the optimal weights when there are too many multiscale features. According to the conclusion in the above experiment, we extract three multiscale features of each image to evaluate the influence of weights. The first weight α_1 is set to a constant number 1, and the square roots of the other two weights α_2 and α_3 range from 0.2 to 5. The recognition accuracies versus weights obtained utilizing MSK-KSVD and MSIK-DL are shown in Fig. 6(a) and (b). The best recognition accuracy (93.01%) of the MSK-KSVD is acquired at $(\sqrt{\alpha_2}, \sqrt{\alpha_3}) = (1, 4)$, which is close to 93.05% acquired with six multiscale features shown in Fig. 5. And the best accuracy of MSIK-DL is 94.36% acquired at $(\sqrt{\alpha_2}, \sqrt{\alpha_3}) = (0.8, 3)$, which is higher than 93.95% obtained using six multiscale features. The results provide an important insight that finding the optimal weights for a small number of multiscale features is better than excessively increasing the number of multiscale features.

4) Dictionary Size: In order to assess the impact of the dictionary size J, we compare the proposed algorithms with some representative DL methods, such as K-SVD, kernel K-SVD, D-KSVD, and LC-KSVD. The dictionary size J ranges from 50 to 300 and all the methods have the same sparsity level $T = 15$. The recognition results obtained using different methods are presented in Fig. 7. It can be seen from the curves that the recognition accuracies of the proposed MSK-KSVD and MSIK-DL report significantly higher than that of the other DL methods, even if the dictionary size is small. The most surprising aspect of the curves is that with successive increases in the dictionary size, the recognition accuracies of these methods have a remarkable improvement except for that of MSIK-DL, which is somewhat counterintuitive. This observation may be explained by the fact that MSIK-DL

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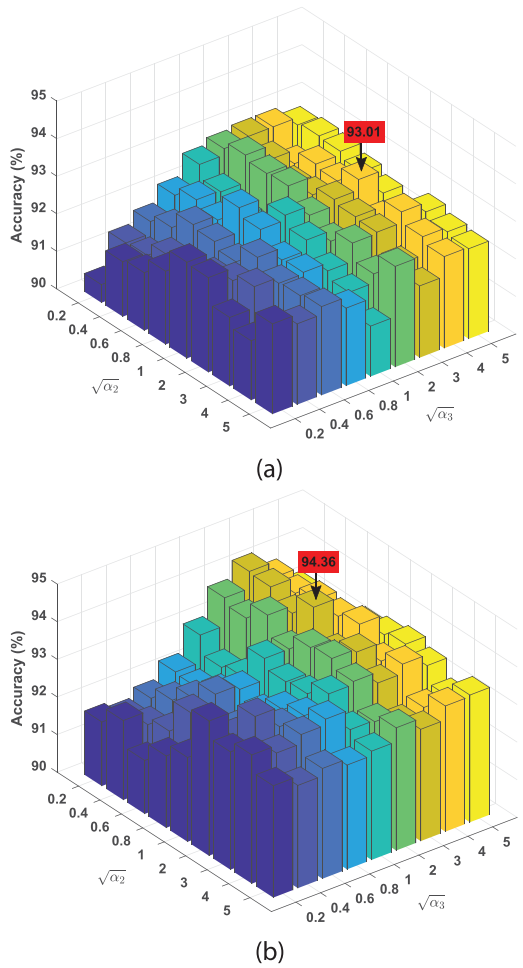


Fig. 6. Recognition accuracy versus weights of different multiscale features using (a) MSK-KSVD and (b) MSIK-DL.

utilizes the stochastic gradient descent to update all atoms of its dictionary at each iteration to find the minimum of the loss function $(1/2)\|\boldsymbol{H} - \boldsymbol{W}\boldsymbol{X}\|_F^2 + (\lambda/2)\|\boldsymbol{W}\|_F^2$ and the values are almost the same with different dictionary size after Lth iteration. Nevertheless, the other optimization methods are based on SVD operation and update the dictionary atom by atom. Each atom is defined as the first column of the orthogonal matrix after each SVD operation. Hence, it can be found intuitively that increasing the number of atoms can decrease the value of loss function, which leads to the improvement of recognition performance relatively.

C. Discrimination Performance

The purpose of this experiment is to evaluate the discrimination performance of the proposed scheme. For a recognition task, reliable discrimination performance is that the sparse codes from the same class are highly correlated with each other and those from different classes are quite independent. Here, we utilize the correlation coefficient [39] to measure it, which is expressed as

$$\rho(\boldsymbol{x}_i, \boldsymbol{x}_j) = \frac{(\boldsymbol{x}_i - \bar{\boldsymbol{x}})^T(\boldsymbol{x}_j - \bar{\boldsymbol{x}})}{\|\boldsymbol{x}_i - \bar{\boldsymbol{x}}\|_2 \|\boldsymbol{x}_j - \bar{\boldsymbol{x}}\|_2} \tag{41}$$

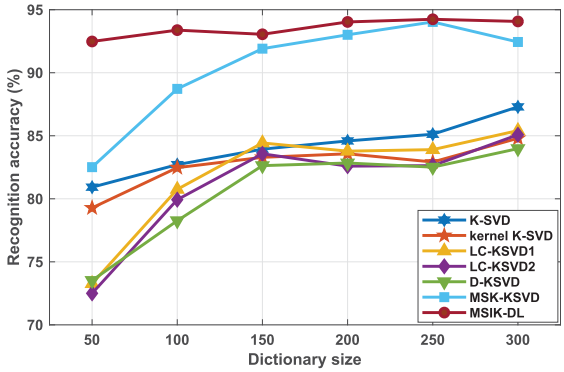


Fig. 7. Recognition accuracy versus number of multiscale feature with different numbers of training samples using MSK-KSVD and MSIK-DL.

where $\bar{\boldsymbol{x}}$ is the mean of the columns of the sparse code matrix $\boldsymbol{X}$. Notably, the correlation coefficient represents the inner product of two normalized vectors. And the larger the absolute value of ρ is, the stronger the correlation of $\boldsymbol{x}_i$ and $\boldsymbol{x}_j$ is. Here, three groups of comparative experiments are conducted: 1) discrimination performance of different methods with the same parameter setting, i.e., dictionary size and sparsity level; 2) discrimination performance of MSK-KSVD and MSIK-DL with various discriminative matrices ($\boldsymbol{H}$, $\boldsymbol{Q}_1$ and $\boldsymbol{Q}_2$); and 3) discrimination performance of MSK-KSVD with different weights β on the discriminative classification error term. We randomly choose 30 samples from each training class for the training stage and the rest training samples are used for validation. We compute the sparse code matrix of the validation samples. Then, the correlation between a validation sample $\boldsymbol{x}_i$ and a category c is defined as

$$\rho_c(\boldsymbol{x}_i) = \frac{1}{N_c} \sum_{h_j = c} \left|\rho(\boldsymbol{x}_i, \boldsymbol{x}_j)\right| \tag{42}$$

where N_c is the number of samples from class c. Fig. 8 displays an example of correlation for all validation samples when $c = 1$ and 10, where the horizontal axis represents validation samples arranged in classes from TG1 to TG10. We can see that the samples from class one and ten have a stronger correlation with classes one and ten, respectively. And what stands out in Fig. 8 is that the discrimination performance of the two proposed algorithms outperforms the other representative DL methods resulting from the consideration of multiscale features, nonlinear situation and label information of SAR images. The K-SVD and kernel K-SVD methods show poor discrimination performance, since these two methods do not utilize label information. From Fig. 8(g) and (h), it is apparent that they show the same discrimination performance, and so are Fig. 8(i) and (j), which demonstrate the Theory 1. In addition, Fig. 8(i), (k), and (l) illustrate that appropriately increasing the weight β on the discriminative classification error term could improve the discrimination performance to some extent.

D. Comparison With Different Methods

We compare the proposed recognition scheme with some classic DL and SR, including linear and nonlinear and

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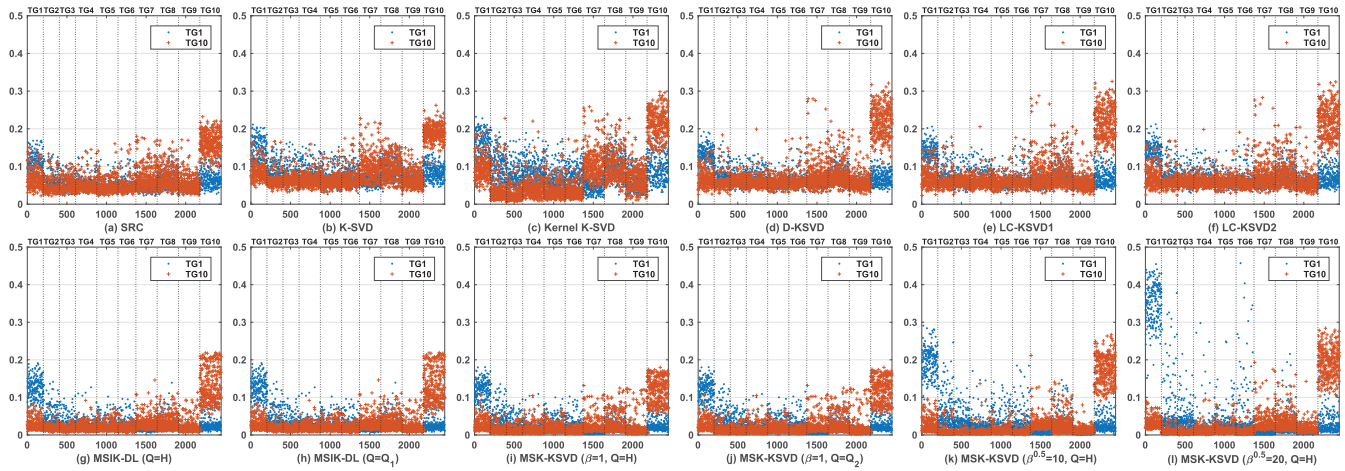


Fig. 8. Discrimination performance of two classes using different DL methods. We select 30 training samples per class to learn the dictionaries and each subfigure shows the mean score of the absolute value of the correlation coefficient for validation samples' sparse codes. (a)–(f) Conventional DL methods' discrimination performance. (g) and (h) Proposed MSIK-DL algorithm's discrimination performance using the label matrix $\boldsymbol{H}$ and the randomly generated matrix $\boldsymbol{Q}_1$ as the discrimination matrices, respectively. (i) and (j) Proposed MSK-KSVD algorithm's discrimination performance using $\boldsymbol{H}$ and $\boldsymbol{Q}_2$ as the discrimination matrices, respectively. (i), (k), and (l) Proposed MSK-KSVD algorithm's discrimination performance with different weights on the discrimination error term.

TABLE IV
SAR TARGET RECOGNITION IN THREE VARIOUS CASES

	Training set size	Testing set size
Case 1	select 30 training samples from each class	use all testing samples from each class
Case 2	select 60 training samples from each class	
Case 3	use all training samples from each class	

supervised and unsupervised methods. Another two conventional classifiers and three state-of-the-art CNN models are also taken into consideration for SAR target recognition. We take into account the small, middle, and large training set size conditions, respectively, which are listed in Table IV. For a fair comparison, all DL methods have the same dictionary size in the three cases, which are set to 200, 400, and 600 when 30, 60, and all training samples per class are selected to constitute the training set. The sparsity level of SRC, K-SVD, LC-KSVD, D-KSVD, kernel K-SVD, and the two proposed algorithms is set to 15. The classification scheme we utilize for SRC, K-SVD, and kernel K-SVD, which are unsupervised DL methods, is to find the minimal reconstruction error. That is, we first learn a subdictionary for each class and then the predicted label is calculated by accessing which class of dictionary could produce minimal reconstruction error. The other comparative methods' recognition schemes are referred to in terms of the corresponding references. We randomly select 30 and 60 samples of each class for training (cases 1 and 2) to evaluate the SAR target recognition performance under small training set size condition and repeat the experiments ten times for each condition with different random selections of training samples.

The mean values of recognition accuracies under SOC shown in Table V demonstrate that our proposed MSK-DL scheme optimized using MSK-KSVD and MSIK-DL outperforms significantly the other competing methods under small training set size condition. There are at least 8% and 5% improvements compared with GoogLeNet in cases 1 and 2,

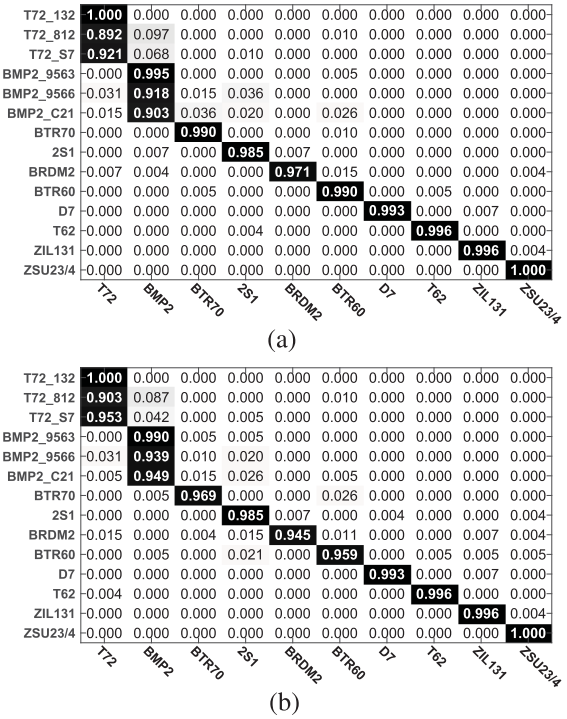
TABLE V
RECOGNITION ACCURACIES OF DIFFERENT METHODS UNDER SOC

Number of train. samp. per class	30 train. samp.	60 train. samp.	all train. samp.
SRC [16]	75.55	87.36	95.04
K-SVD [18]	73.21	86.54	96.41
LC-KSVD1[2] [20]	72.66	85.75	95.13
LC-KSVD2[2] [20]	71.02	84.06	93.91
D-KSVD [19]	71.19	84.01	94.19
Kernel K-SVD [23]	72.44	84.28	94.10
FDDL[3] [21]	74.11	85.79	94.07
LRSDL[3] [40]	74.76	87.18	95.54
LRSR[4] [41]	77.32	83.99	88.45
PCA+SVM [42]	73.71	88.14	94.32
CR-based k-NN[5] [7]	77.01	86.31	89.82
GoogLeNet[6] [43]	78.83	88.88	96.07
ResNet-50[6] [44]	75.82	86.64	**97.91**
ResNeXt-101[6] [45]	78.17	87.17	96.78
MSK-KSVD	**86.59**	**93.96**	97.13
MSIK-DL	**87.13**	**94.07**	97.22

respectively, since the proposed MSK-DL scheme takes into consideration an SAR image's multiscale features, nonlinear situation, and discrimination performance together. When all training samples are trained (case 3), our methods show slightly lower recognition accuracy than ResNet-50. However, the training time of deep learning models is much longer than DL methods. Fig. 9(a) and (b) displays the confusion matrices of MSK-KSVD and MSIK-DL in case 3. It can be found that the subclasses SN_812 and SN_S7 of T72 and SN_9566 and SN_C21 of BMP2 have relatively lower recognition accuracies than the other classes. This observation may be explained by the fact that the training set has no samples of the corresponding subclasses.

[2]downloaded from http://users.umiacs.umd.edu/zhuolin/
[3]downloaded from https://github.com/tiepvupsu/DICTOL
[4]downloaded from https://ganggangdong.github.io/homepage/
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[6]downloaded from https://github.com/pytorch/vision/tree/master/torchvision

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(a)

(b)

Fig. 9. Confusion matrices obtained using MSK-KSVD and MSIK-DL under SOC. (a) MSK-KSVD. (b) MSIK-DL.

TABLE VI
CLASSES UNDER EOCS

Target		TG4	TG5	TG10	TG1	Total
		2S1	BRDM2	ZSU23/4	T72 (SN_A64)	
Train	17°	299	298	299	299	1195
Test	15°	274	274	274	274	1096
	30°	288	287	288	288	1151

In addition, we use the MSTAR data set under extended operating conditions (EOCs) to assess the generalization ability of the proposed scheme, where the samples of four classes acquired at depression angle 17° are utilized for training and those acquired at depression angles 15° and 30° are exploited in the testing stage. The size of training and testing sets is listed in Table VI. The aforementioned SVM, k-NN, and DL methods are taken into account under EOC to evaluate the robustness to depression angles. The values of parameters are set as those under SOC. The recognition results are shown in Table VII. It can be seen that the proposed method still maintains the highest classification accuracies in most cases, though its accuracies are slightly lower than CR-based k-NN (89.75%) at depression angle 30° in case 1 and LRSR (99.54%) at depression angle 15° in case 3. It can also be concluded that SRC, K-SVD, kernel K-SVD, and SVM combined with PCA are sensitive to depression angles while the proposed scheme shows its decent robustness to depression angles.

E. Robustness to Noise

In this part, in order to evaluate the performance of the proposed method in robustness to noise, we add some multiplicative speckle noise with varying signal-to-noise ratios (SNRs) into each testing SAR image and compare the proposed scheme with some representative DL methods. The speckle noisy SAR image $\hat{\boldsymbol{I}}$ is obtained according to

$$\hat{\boldsymbol{I}} = \boldsymbol{I} + \boldsymbol{N} \circ \boldsymbol{I} \tag{43}$$

where $\boldsymbol{N}$ is uniformly distributed random noise with mean 0 and variance υ, and $\circ$ represents the element-wise product operator. Denote $P(\boldsymbol{I})$ as the energy of an image $\boldsymbol{I}$. Then, SNR is defined as

$$\text{SNR} = 10\log_{10}\frac{P(\boldsymbol{I})}{P(\boldsymbol{N} \circ \boldsymbol{I})}. \tag{44}$$

Here, we choose case 2 under SOC as an instance and obtain the average accuracies after ten runs with random selections of training samples. The recognition results are shown in Table VIII, which illustrates that the recognition accuracies of all methods increase with higher SNRs, however, the proposed scheme shows the highest accuracies within a range of −5–20 dB. And the recognition accuracy is still over 65% when SNR is −5 dB, whereas the accuracies of the other methods are lower than 55%. This means that the proposed scheme is more robust to noise.

TABLE VII
RECOGNITION ACCURACIES OF DIFFERENT METHODS UNDER EOC

Number of train. samp. per class	30 train. samp.		60 train. samp.		all train. samp.	
Depression angle	15°	30°	15°	30°	15°	30°
SRC	87.92	82.86	94.90	91.02	98.63	97.83
K-SVD	86.06	78.59	94.47	87.29	98.63	96.52
LC-KSVD1	85.03	82.44	93.80	90.60	98.54	97.74
LC-KSVD2	83.28	83.57	92.96	91.03	98.18	98.00
D-KSVD	83.41	83.65	93.10	91.17	98.72	97.65
Kernel K-SVD	85.53	75.86	93.69	84.67	98.45	92.88
FDDL	86.56	84.71	90.03	92.98	98.54	**99.04**
LRSDL	86.06	82.21	93.59	93.39	98.63	98.61
LRSR	91.41	83.75	97.35	91.07	**99.54**	95.31
PCA+SVM	75.84	57.88	89.46	72.64	98.27	82.19
CR-based k-NN	89.00	**89.75**	95.20	94.70	99.18	95.66
MSK-KSVD	**91.68**	88.06	**97.60**	**95.54**	99.18	**99.04**
MSIK-DL	**92.72**	89.24	**97.78**	**95.93**	99.27	98.96

TABLE VIII
RECOGNITION ACCURACIES OF DIFFERENT METHODS WITH VARYING SNRS UNDER SOC

SNR (dB)	-5	0	5	10	15	20
SRC	50.18	72.32	82.63	86.18	86.95	87.18
K-SVD	41.71	70.57	84.31	86.85	86.94	86.80
LC-KSVD1	51.53	73.33	82.91	85.72	86.17	86.08
LC-KSVD2	54.47	75.40	83.22	85.19	85.07	84.69
D-KSVD	53.10	74.20	82.60	84.61	84.54	84.33
Kernel K-SVD	36.95	67.06	81.66	84.61	84.97	84.76
MSK-KSVD	**65.92**	**82.43**	**89.81**	**93.16**	**94.03**	**94.38**
MSIK-DL	**66.19**	**82.54**	**90.08**	**93.52**	**94.41**	**94.72**

VII. CONCLUSION

This article introduces a nonlinear DL strategy for SAR ATR. Unlike the previous works, which are single linear or nonlinear DL methods, we propose to utilize Frost filters with different parameters to extract multiscale features of an

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SAR image to solve the small training set size and speckle noise problems. In addition, nonlinear situation is also considered when reconstructing each scale feature, thus we apply a nonlinear function to map these multiscale features to a nonlinear Hilbert space such that multinonlinear dictionaries are learned. The discriminative classification error term is exploited to learn a classifier and make the sparse codes more discriminative, that is, sparse codes from the same class are similar and those from different classes are more independent. Multiple experiments are performed to evaluate the two proposed optimization algorithms on the MSTAR data set, from which the following conclusions can be obtained.

1) The two proposed MSK-KSVD and MSIK-DL show decent convergence performance. The learning rate of MSIK-DL should be selected prudently to avoid slow convergence rate or local oscillation.
2) The recognition performance improves remarkably when increasing the number of multiscale features and the dictionary size.
3) We can select relative optimal dimension of each scale feature and weights on each reconstruction error term to improve the recognition accuracy.
4) The form of the discriminative error term does not affect the discrimination of sparse codes.

The experimental results also demonstrate that the two optimization methods have similar recognition accuracies. There are several possible explanations for the small differences, that is: 1) using ℓ_1-norm to replace ℓ_0-norm in MSIK-DL; 2) placing the nonlinear reconstruction error terms into the constraint term leading to the removal of the weight on the classification error term in MSIK-DL; and 3) the different forms of update at each iteration. The proposed strategy outperforms the other representative methods and shows a strong antinoise performance.

APPENDIX

A. Proof of Theorem 1

For every discriminative matrix $\boldsymbol{Q}$, which satisfies the constraint (11), it is not difficult to obtain that $\boldsymbol{q}_{i,1} = \boldsymbol{q}_{i,2} = \boldsymbol{q}_{i,N_i} = \boldsymbol{q}_i$, and $\|\boldsymbol{q}_i\|_2 = 1$, that is

$$\boldsymbol{Q} = \left[\underbrace{\boldsymbol{q}_1, \ldots, \boldsymbol{q}_1}_{N_1}, \underbrace{\boldsymbol{q}_2, \ldots, \boldsymbol{q}_2}_{N_2}, \ldots, \underbrace{\boldsymbol{q}_M, \ldots, \boldsymbol{q}_M}_{N_M}\right]. \quad (45)$$

Define an orthogonal matrix

$$\boldsymbol{Q}^\star = \left[\boldsymbol{q}_1, \ldots, \boldsymbol{q}_M, \boldsymbol{q}_{M+1}, \ldots, \boldsymbol{q}_J\right] \in \mathbb{R}^{J\times J} \quad (46)$$

and we have $\|\boldsymbol{Q}^{\star T}\boldsymbol{Z}\|_F^2 = \|\boldsymbol{Z}\|_F^2$ for any matrix $\boldsymbol{Z}$. Hence, the objective function of (12) is equivalent to

$$\begin{aligned}
&\sum_{k=1}^{K} \alpha_k \|\Phi(\boldsymbol{P}_k) - \boldsymbol{D}_k\boldsymbol{X}\|_F^2 + \gamma \|\boldsymbol{Q} - \boldsymbol{G}\boldsymbol{X}\|_F^2 \\
&\quad = \sum_{k=1}^{K} \alpha_k \|\Phi(\boldsymbol{P}_k) - \boldsymbol{D}_k\boldsymbol{X}\|_F^2 + \gamma \left\|\boldsymbol{Q}^{\star T}(\boldsymbol{Q} - \boldsymbol{G}\boldsymbol{X})\right\|_F^2 \\
&\quad = \sum_{k=1}^{K} \alpha_k \|\Phi(\boldsymbol{P}_k) - \boldsymbol{D}_k\boldsymbol{X}\|_F^2 + \gamma \|\widetilde{\boldsymbol{Q}} - \widetilde{\boldsymbol{G}}\boldsymbol{X}\|_F^2 \quad (47)
\end{aligned}$$

where $\widetilde{\boldsymbol{Q}} = \boldsymbol{Q}^{\star T}\boldsymbol{Q} = [\boldsymbol{H}^T, \boldsymbol{0}^T]^T$ and $\widetilde{\boldsymbol{G}} = \boldsymbol{Q}^{\star T}\boldsymbol{G}$. As a result, the reformulation of problem (12) is

$$\begin{aligned}
&\min_{\boldsymbol{D}_k, \widetilde{\boldsymbol{G}}, \boldsymbol{X}} \sum_{k=1}^{K} \alpha_k \|\Phi(\boldsymbol{P}_k) - \boldsymbol{D}_k\boldsymbol{X}\|_F^2 + \gamma \left\|\widetilde{\boldsymbol{Q}} - \widetilde{\boldsymbol{G}}\boldsymbol{X}\right\|_F^2 \\
&\text{s.t. } \|\boldsymbol{x}_i\|_0 \leqslant T, \quad i = 1, 2, \ldots, N. \quad (48)
\end{aligned}$$

Suppose that $\langle \boldsymbol{D}_k^*, \boldsymbol{W}^* \rangle$ is the optimal solution of the following optimization problem using the proposed MSK-KSVD or MSIK-DL algorithms:

$$\begin{aligned}
&\min_{\boldsymbol{D}_k, \boldsymbol{W}, \boldsymbol{X}} \sum_{k=1}^{K} \alpha_k \|\Phi(\boldsymbol{P}_k) - \boldsymbol{D}_k\boldsymbol{X}\|_F^2 + \beta \|\boldsymbol{H} - \boldsymbol{W}\boldsymbol{X}\|_F^2 \\
&\text{s.t. } \|\boldsymbol{x}_i\|_0 \leqslant T, \quad i = 1, 2, \ldots, N. \quad (49)
\end{aligned}$$

The following is to prove that the optimal solution of problem (48) is $\langle \widetilde{\boldsymbol{D}}_k^*, \widetilde{\boldsymbol{G}}^* \rangle$ when $\gamma = \beta$, where $\widetilde{\boldsymbol{D}}_k^* = \boldsymbol{D}_k^*$ and $\widetilde{\boldsymbol{G}}^* = [\boldsymbol{W}^{*T}, \boldsymbol{0}^T]$, such that the sought dictionaries are the same and have the same discrimination performance for any designed matrix $\boldsymbol{Q}$ which satisfies the constraint (11).

We utilize mathematical induction for proof:

Initialization: With a similar transformation (14)–(17), we use one-hot vector to initialize $\{\widetilde{\boldsymbol{A}}_k^{(0)}\}$ and $\{\boldsymbol{A}_k^{(0)}\}$. $\widetilde{\boldsymbol{G}}^{(0)}$ is obtained according to

$$\begin{aligned}
\widetilde{\boldsymbol{G}}^{(0)} &= \boldsymbol{Q}^{\star T}\boldsymbol{G}^{(0)} = \boldsymbol{Q}^{\star T}\boldsymbol{Q}\boldsymbol{X}^T\left(\boldsymbol{X}\boldsymbol{X}^T + \lambda\boldsymbol{I}\right)^{-1} \\
&= \begin{bmatrix}\boldsymbol{H}\\ \boldsymbol{0}\end{bmatrix}\boldsymbol{X}^T\left(\boldsymbol{X}\boldsymbol{X}^T + \lambda\boldsymbol{I}\right)^{-1} = \begin{bmatrix}\boldsymbol{W}^{(0)}\\ \boldsymbol{0}\end{bmatrix}. \quad (50)
\end{aligned}$$

Assume that

$$\widetilde{\boldsymbol{A}}_k^{(l)} = \boldsymbol{A}_k^{(l)}, \quad \widetilde{\boldsymbol{G}}^{(l)} = \begin{bmatrix}\boldsymbol{W}^{(l)}\\ \boldsymbol{0}\end{bmatrix}. \quad (51)$$

The next step is to prove

$$\widetilde{\boldsymbol{A}}_k^{(l+1)} = \boldsymbol{A}_k^{(l+1)}, \quad \widetilde{\boldsymbol{G}}^{(l+1)} = \begin{bmatrix}\boldsymbol{W}^{(l+1)}\\ \boldsymbol{0}\end{bmatrix}. \quad (52)$$

1) Equivalence Using MSK-KSVD: This algorithm is divided into two steps.

Step One (Sparse Coding): According to (51), we can calculate the sparse code matrix using KOMP algorithm and obtain

$$\widetilde{\boldsymbol{X}}^{(l+1)} = \boldsymbol{X}^{(l+1)}. \quad (53)$$

Step Two (Dictionaries Update): According to the process of MSK-KSVD introduced in Section IV-B and with the following substitution:

$$\widetilde{\Psi}(\boldsymbol{P}) = \left[\Psi(\boldsymbol{P})^T, \sqrt{\gamma}\widetilde{\boldsymbol{Q}}^T\right]^T \quad (54a)$$

$$\widetilde{\Gamma}(\boldsymbol{P}) = \operatorname{diag}\left(\Gamma(\boldsymbol{P}), \widetilde{\boldsymbol{I}}\right) \quad (54b)$$

$$\widetilde{\boldsymbol{B}}' = \left[\widetilde{\boldsymbol{B}}^T, \sqrt{\gamma}\widetilde{\boldsymbol{G}}^T\right]^T \in \mathbb{R}^{(N\times K+J)\times J} \quad (54c)$$

the Gram matrix is

$$\begin{aligned}
\widetilde{\boldsymbol{E}}^T\widetilde{\boldsymbol{E}} &= \widetilde{\Psi}(\boldsymbol{P})^T\widetilde{\Psi}(\boldsymbol{P}) + \left(\sum_{j\neq n}\widetilde{\boldsymbol{b}}_j'\boldsymbol{x}^j\right)^T\widetilde{\Gamma}(\boldsymbol{P})^T\widetilde{\Gamma}(\boldsymbol{P})\sum_{j\neq n}\widetilde{\boldsymbol{b}}_j'\boldsymbol{x}^j \\
&\quad - \left(\sum_{j\neq n}\widetilde{\boldsymbol{b}}_j'\boldsymbol{x}^j\right)^T\widetilde{\Gamma}(\boldsymbol{P})^T\widetilde{\Psi}(\boldsymbol{P}) - \widetilde{\Psi}(\boldsymbol{P})^T\widetilde{\Gamma}(\boldsymbol{P})\sum_{j\neq n}\widetilde{\boldsymbol{b}}_j'\boldsymbol{x}^j \\
&= \boldsymbol{E}^T\boldsymbol{E}. \quad (55)
\end{aligned}$$

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Therefore, after the lth iteration, we have $\widetilde{\boldsymbol{v}}_1 = \boldsymbol{v}_1$ and $\widetilde{\sigma}_1 = \sigma_1$. In terms of (30), we have

$$\begin{aligned}\left(\widetilde{\boldsymbol{a}}_k\right)_n^{(l+1)} &= \widetilde{\sigma}_1^{-1}\left(\boldsymbol{I} - \sum_{j\neq n}\left(\widetilde{\boldsymbol{a}}_k\right)_j^{(l)}\left(\widetilde{\boldsymbol{x}}^j\right)^{(l+1)}\right)\boldsymbol{\Omega}_n\widetilde{\boldsymbol{v}}_1 \\ &= \left(\boldsymbol{a}_k\right)_n^{(l+1)} \qquad (56)\\ \widetilde{\boldsymbol{g}}_n^{(l+1)} &= \widetilde{\sigma}_1^{-1}\left(\widetilde{\boldsymbol{Q}} - \sum_{j\neq n}\widetilde{\boldsymbol{g}}_j^{(l)}\left(\widetilde{\boldsymbol{x}}^j\right)^{(l+1)}\right)\boldsymbol{\Omega}_n\widetilde{\boldsymbol{v}}_1 \\ &= \widetilde{\sigma}_1^{-1}\left(\begin{bmatrix}\boldsymbol{H}\\ \boldsymbol{0}\end{bmatrix} - \sum_{j\neq n}\begin{bmatrix}\boldsymbol{w}_j^{(l)}\\ \boldsymbol{0}\end{bmatrix}\left(\widetilde{\boldsymbol{x}}^j\right)^{(l+1)}\right)\boldsymbol{\Omega}_n\widetilde{\boldsymbol{v}}_1 \\ &= \begin{bmatrix}\boldsymbol{w}_n^{(l+1)}\\ \boldsymbol{0}\end{bmatrix}. \qquad (57)\end{aligned}$$

2) Equivalence Using MSIK-DL: For this incremental method, we just need to prove that the value of the gradient is equivalent. It is not difficult to compute that $\widetilde{\boldsymbol{\varphi}}^{(l)} = \boldsymbol{\varphi}^{(l)}$. Therefore, according to (34a), we have

$$\frac{\partial \mathcal{L}_i}{\partial \widetilde{\boldsymbol{B}}} = \frac{\partial \mathcal{L}_i}{\partial \boldsymbol{B}}. \qquad (58)$$

The gradient of $\mathcal{L}_i$ with respect to $\widetilde{\boldsymbol{G}}$ is

$$\begin{aligned}\frac{\partial \mathcal{L}_i}{\partial \widetilde{\boldsymbol{G}}} &= \left(\widetilde{\boldsymbol{G}}^{(l)}\boldsymbol{x}_i - \widetilde{\boldsymbol{q}}_i\right)\boldsymbol{x}_i^T + \lambda\widetilde{\boldsymbol{G}}^{(l)} \\ &= \left(\begin{bmatrix}\boldsymbol{W}^{(l)}\\ \boldsymbol{0}\end{bmatrix}\boldsymbol{x}_i - \begin{bmatrix}\boldsymbol{h}_i\\ \boldsymbol{0}\end{bmatrix}\right)\boldsymbol{x}_i^T + \lambda\begin{bmatrix}\boldsymbol{W}^{(l)}\\ \boldsymbol{0}\end{bmatrix} \\ &= \begin{bmatrix}\frac{\partial \mathcal{L}_i}{\partial \boldsymbol{W}}\\ \boldsymbol{0}\end{bmatrix}. \qquad (59)\end{aligned}$$

Finally, in terms of the gradient decent method, we can obtain

$$\widetilde{\boldsymbol{A}}_k^{(l+1)} = \boldsymbol{A}_k^{(l+1)},\ \widetilde{\boldsymbol{G}}^{(l+1)} = \begin{bmatrix}\boldsymbol{W}^{(l+1)}\\ \boldsymbol{0}\end{bmatrix}. \qquad (60)$$

B. Derivation of MSIK-DL

According to (33), we can obtain

$$\begin{aligned}&\frac{\partial\left[\boldsymbol{B}^T\Gamma(\boldsymbol{P})^T\left(\Gamma(\boldsymbol{P})\boldsymbol{B}\boldsymbol{x}_i - \Psi\left(\boldsymbol{p}^i\right)\right)\right]}{\partial \boldsymbol{B}_{pq}} = 0 \qquad (61)\\ \Rightarrow\ &\frac{\partial\left[\boldsymbol{B}^T\Gamma(\boldsymbol{P})^T\Gamma(\boldsymbol{P})\boldsymbol{B}\right]}{\partial \boldsymbol{B}_{pq}}\boldsymbol{x}_i - \frac{\partial\left[\boldsymbol{B}^T\Gamma(\boldsymbol{P})^T\Psi\left(\boldsymbol{p}^i\right)\right]}{\partial \boldsymbol{B}_{pq}} \\ &+ \boldsymbol{B}^T\Gamma(\boldsymbol{P})^T\Gamma(\boldsymbol{P})\boldsymbol{B}\frac{\partial \boldsymbol{x}_i}{\partial \boldsymbol{B}_{pq}} = 0 \qquad (62)\\ &\frac{\partial \mathcal{L}_i}{\partial \boldsymbol{B}_{pq}} = \frac{\partial \mathcal{L}_i}{\partial \boldsymbol{x}_i^T}\frac{\partial \boldsymbol{x}_i}{\partial \boldsymbol{B}_{pq}} = \frac{\partial \mathcal{L}_i}{\partial \boldsymbol{x}_i^T}\left(\boldsymbol{B}^T\Gamma(\boldsymbol{P})^T\Gamma(\boldsymbol{P})\boldsymbol{B}\right)^{-1} \\ \Rightarrow\ &\left[\frac{\partial\left(\boldsymbol{B}^T\Gamma(\boldsymbol{P})^T\Psi\left(\boldsymbol{p}^i\right)\right)}{\partial \boldsymbol{B}_{pq}} - \frac{\partial\left(\boldsymbol{B}^T\Gamma(\boldsymbol{P})^T\Gamma(\boldsymbol{P})\boldsymbol{B}\right)}{\partial \boldsymbol{B}_{pq}}\boldsymbol{x}_i\right]. \qquad (63)\end{aligned}$$

Denote $\boldsymbol{\mathcal{K}} = [\boldsymbol{\mathcal{K}}_{ij}] = \Gamma(\boldsymbol{P})^T\Gamma(\boldsymbol{P}) \in \mathbb{R}^{NK\times NK}$, $\boldsymbol{\kappa} = [\boldsymbol{\kappa}_i] = \Gamma(\boldsymbol{P})^T\Psi(\boldsymbol{p}^i) \in \mathbb{R}^{NK\times 1}$ and $\widehat{\boldsymbol{\varphi}}^T = (\partial \mathcal{L}_i/\partial \boldsymbol{x}_i^T)(\boldsymbol{B}^T\Gamma(\boldsymbol{P})^T\Gamma(\boldsymbol{P})\boldsymbol{B})^{-1} \in \mathbb{R}^{1\times J}$. Then, the right side of (63) can be divided into two parts. The first part is

$$\widehat{\boldsymbol{\varphi}}^T\frac{\partial \boldsymbol{B}^T}{\partial \boldsymbol{B}_{pq}}\boldsymbol{\kappa} = \left[\widehat{\boldsymbol{\varphi}}_1, \widehat{\boldsymbol{\varphi}}_2, \ldots, \widehat{\boldsymbol{\varphi}}_J\right]\begin{bmatrix}0\\ \vdots\\ \boldsymbol{\kappa}_p\\ \vdots\\ 0\end{bmatrix}_{J\times 1} = \boldsymbol{\kappa}_p\widehat{\boldsymbol{\varphi}}_q \qquad (64)$$

where $\boldsymbol{\kappa}_p$ is the qth element of the vector $[0, \ldots, \boldsymbol{\kappa}_p, \ldots, 0]^T$.

The second part is

$$\begin{aligned}\widehat{\boldsymbol{\varphi}}^T\frac{\partial\left(\boldsymbol{B}^T\boldsymbol{\mathcal{K}}\boldsymbol{B}\right)}{\partial \boldsymbol{B}_{pq}}\boldsymbol{x}_i &= \widehat{\boldsymbol{\varphi}}^T\frac{\partial \boldsymbol{B}^T}{\partial \boldsymbol{B}_{pq}}\boldsymbol{\mathcal{K}}\boldsymbol{B}\boldsymbol{x}_i + \widehat{\boldsymbol{\varphi}}^T\boldsymbol{B}^T\boldsymbol{\mathcal{K}}\frac{\partial \boldsymbol{B}}{\partial \boldsymbol{B}_{pq}}\boldsymbol{x}_i \\ &= \left(\boldsymbol{\mathcal{K}}\boldsymbol{B}\boldsymbol{x}_i\right)_p\widehat{\boldsymbol{\varphi}}_q + \left(\widehat{\boldsymbol{\varphi}}^T\boldsymbol{B}^T\boldsymbol{\mathcal{K}}\right)_p\left(\boldsymbol{x}_i\right)_q. \qquad (65)\end{aligned}$$

Therefore, we obtain

$$\begin{aligned}&\frac{\partial \mathcal{L}_i}{\partial \boldsymbol{B}_{pq}} = \boldsymbol{\kappa}_p\widehat{\boldsymbol{\varphi}}_q - \left(\boldsymbol{\mathcal{K}}\boldsymbol{B}\boldsymbol{x}_i\right)_p\widehat{\boldsymbol{\varphi}}_q - \left(\widehat{\boldsymbol{\varphi}}^T\boldsymbol{B}^T\boldsymbol{\mathcal{K}}\right)_p\left(\boldsymbol{x}_i\right)_q \qquad (66)\\ \Rightarrow\ &\frac{\partial \mathcal{L}_i}{\partial \boldsymbol{B}} = \boldsymbol{\kappa}\widehat{\boldsymbol{\varphi}}^T - \left(\boldsymbol{\mathcal{K}}\boldsymbol{B}\boldsymbol{x}_i\right)\widehat{\boldsymbol{\varphi}}^T - \left(\boldsymbol{\mathcal{K}}^T\boldsymbol{B}\widehat{\boldsymbol{\varphi}}\right)\boldsymbol{x}_i^T. \qquad (67)\end{aligned}$$

Notably, there is no definition of $\partial[-\gamma\,\mathrm{sgn}(\boldsymbol{x}_i)]/\partial \boldsymbol{B}_{pq}$ for $(\boldsymbol{x}_i)_l = 0$ in (33). Let Λ be the index set of nonzero values of $\boldsymbol{x}_i$ and $\bar{\Lambda}$ be the index set of zero values of $\boldsymbol{x}_i$. Denote $\boldsymbol{\varphi}_\Lambda = \widehat{\boldsymbol{\varphi}}_\Lambda$ and $\boldsymbol{\varphi}_{\bar{\Lambda}} = 0$. As a result, the gradients of $\mathcal{L}_i$ with respect to $\boldsymbol{B}$ is

$$\frac{\partial \mathcal{L}_i}{\partial \boldsymbol{B}} = -\left(\boldsymbol{\mathcal{K}}^T\boldsymbol{B}\boldsymbol{\varphi}\right)\boldsymbol{x}_i^T + \left(\boldsymbol{\kappa} - \boldsymbol{\mathcal{K}}\boldsymbol{B}\boldsymbol{x}_i\right)\boldsymbol{\varphi}^T. \qquad (68)$$

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Lei Tao (Student Member, IEEE) received the B.S. degree in information engineering from Southeast University, Nanjing, China, in 2016. He is currently pursuing the Ph.D. degree with the School of Electronic Information and Electrical Engineering, Shanghai Jiao Tong University, Shanghai, China.

His research interests include machine learning, image processing, and radar automatic target recognition.



Xue Jiang (Senior Member, IEEE) received the M.S. degree in acoustics and signal processing from the Chinese Academy of Sciences, Beijing, China, in 2007, and the Ph.D. degree in electrical engineering from McMaster University, Hamilton, ON, Canada, in 2014.

From 2007 to 2010, she was a Research Engineer with Sunplus Inc., Beijing. She held a post-doctoral fellow position at McMaster University from 2014 to 2015 and Arizona State University, Tempe, AZ, USA, from 2015 to 2016. She joined the School of Electronic Information and Electrical Engineering, Shanghai Jiao Tong University, Shanghai, China, in 2016, where she is currently an Associate Professor. Her research interests include array processing, robust beamforming, sparse and low-rank learning, data science, and numerical optimization, with applications to radar, sonar, and remote sensing.

Dr. Jiang was one of the finalists of the Best Student Paper Award of the IEEE 2013 Radar Conference. She received the McMaster University International Excellence Award in 2012 and 2013. She is also an Associate Editor of the IEEE TRANSACTIONS ON SIGNAL PROCESSING. She serves as the Editorial Board Member for *IET Radar, Sonar & Navigation*.

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Xingzhao Liu (Member, IEEE) received the B.S. and M.S. degrees from the Harbin Institute of Technology, Harbin, China, in 1984 and 1992, respectively, and the Ph.D. degree from the University of Tokushima, Tokushima, Japan, in 1995, all in electrical engineering.

From 1984 to 1998, he was an Assistant Professor, an Associate Professor, and a Professor with the Harbin Institute of Technology. Since 1998, he has been a Professor with Shanghai Jiao Tong University, Shanghai, China. His research interests include high-frequency radar and synthetic aperture radar signal processing.

Zhou Li received the Ph.D. degree in signal and information processing from the School of Electronics and Information Engineering, Beihang University, Beijing, China, in 2014.

He is currently an Associate Scientist with the Beijing Institute of Remote Sensing Information, Beijing. His research interests include synthetic aperture radar signal processing and high-resolution image formation and analysis.



Zhixin Zhou received the B.S. degree in communication engineering from the Electronic Engineering Institute of PLA, Hefei, China, in 1986, and the M.S. and Ph.D. degrees in electronic engineering from the Harbin Institute of Technology, Harbin, China, in 1989 and 1997, respectively.

He was elected to be a member of the Chinese Academy of Sciences, Beijing, China, in 2015. He is currently the President of the Space Engineering University, Beijing. His research interests include computer vision, pattern recognition, and remote sensing.

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